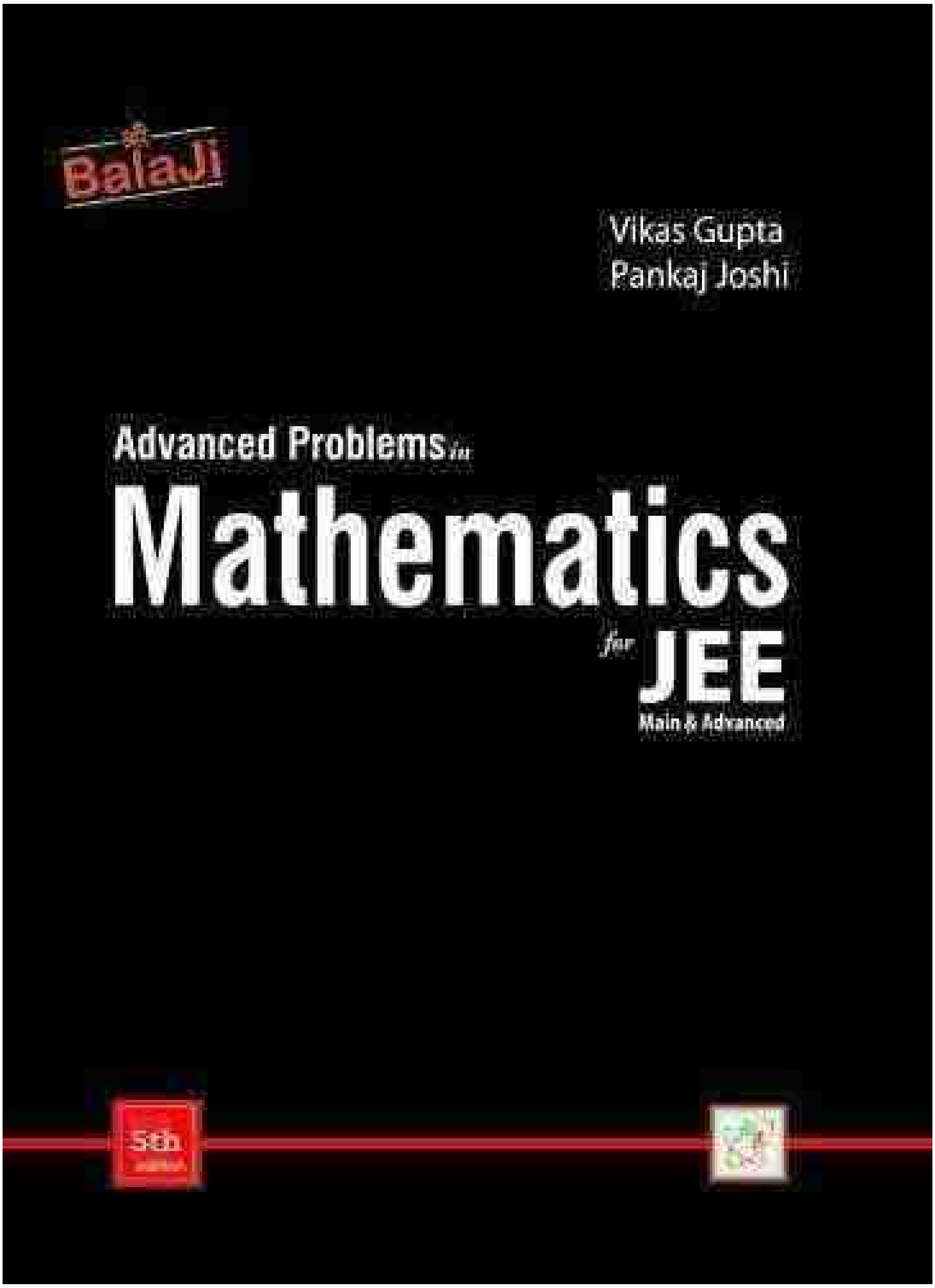
Balaji

Advanced Problems in Mathematics Chapter 1 to 9

for IIT JEE Main and Advanced

by

Vikas Gupta and Pankaj Joshi



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Advanced Problems *in*

MATHEMATICS

for

JEE (MAIN & ADVANCED)

by :

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Chapter 1 – Function

1 FUNCTION

Exercise-1 : Single Choice Problems

1. Range of the function $f(x) = \log_2(2 - \log_{\sqrt{2}}(16\sin^2 x + 1))$ is :

(a) $[0,1]$ (b) $(-\infty, 1]$ (c) $[-1,1]$ (d) $(-\infty, \infty)$

2. The value of a and b for which $|e^{|x-b|} - a| = 2$, has four distinct solutions, are :

(a) $a \in (-3, \infty), b = 0$ (b) $a \in (2, \infty), b = 0$ (c) $a \in (3, \infty), b \in R$ (d) $a \in (2, \infty), b = a$

3. The range of the function :

$$f(x) = \tan^{-1} x + \frac{1}{2}\sin^{-1} x$$

(a) $(-\pi/2, \pi/2)$ (b) $[-\pi/2, \pi/2] - \{0\}$ (c) $[-\pi/2, \pi/2]$ (d) $(-3\pi/4, 3\pi/4)$

4. Find the number of real ordered pair(s) (x, y) for which :

$16^{x^2+y} + 16^{x+y^2} = 1$

(a) 0 (b) 1 (c) 2 (d) 3

5. The complete range of values of 'a' such that $\left(\frac{1}{2}\right)^{|x|} = x^2 - a$ is satisfied for maximum number of values of x is :

(a) $(-\infty, -1)$ (b) $(-\infty, \infty)$ (c) $(-1, 1)$ (d) $(-1, \infty)$

6. For a real number x, let $[x]$ denotes the greatest integer less than or equal to x. Let $f : R \to R$ be defined by $f(x) = 2x + [x] + \sin x \cos x$. Then f is :

(a) One-one but not onto (b) Onto but not one-one

(c) Both one-one and onto (d) Neither one-one nor onto

7. The maximum value of $\sec^{-1}\left(\frac{7 - 5(x^2 + 3)}{2(x^2 + 2)}\right)$ is :

(a) $\frac{5\pi}{6}$ (b) $\frac{5\pi}{12}$ (c) $\frac{7\pi}{12}$ (d) $\frac{2\pi}{3}$

8. Number of ordered pair (a, b) from the set $A = \{1, 2, 3, 4, 5\}$ so that the function $f(x) = \frac{x^3}{3} + \frac{a}{2}x^2 + bx + 10$ is an injective mapping $\forall\, x \in R$:

(a) 13 (b) 14 (c) 15 (d) 16

9. Let A be the greatest value of the function $f(x) = \log_x [x]$, (where $[\cdot]$ denotes greatest integer function) and B be the least value of the function $g(x) = |\sin x| + |\cos x|$, then :

(a) $A > B$ (b) $A < B$ (c) $A = B$ (d) $2A + B = 4$

10. Let $A = [a, \infty)$ denotes domain, then $f : [a, \infty) \to B$, $f(x) = 2x^3 - 3x^2 + 6$ will have an inverse for the smallest real value of a, if :

(a) $a = 1, B = [5, \infty)$ (b) $a = 2, B = [10, \infty)$ (c) $a = 0, B = [6, \infty)$ (d) $a = -1, B = [1, \infty)$

11. Solution of the inequation $\{x\}(\{x\} - 1)(\{x\} + 2) \geq 0$
(where $\{\cdot\}$ denotes fractional part function) is :

(a) $x \in (-2, 1)$ (b) $x \in I$ (I denote set of integers)
(c) $x \in [0, 1)$ (d) $x \in [-2, 0)$

12. Let $f(x), g(x)$ be two real valued functions then the function $h(x) = 2\max\{f(x) - g(x), 0\}$ is equal to :

(a) $f(x) - g(x) - |g(x) - f(x)|$ (b) $f(x) + g(x) - |g(x) - f(x)|$
(c) $f(x) - g(x) + |g(x) - f(x)|$ (d) $f(x) + g(x) + |g(x) - f(x)|$

13. Let $R = \{(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)\}$ be a relation on the set $A = \{1, 2, 3, 4\}$. The relation R is :

(a) a function (b) reflexive (c) not symmetric (d) transitive

14. The true set of values of 'K' for which $\sin^{-1}\left(\frac{1}{1 + \sin^2 x}\right) = \frac{K\pi}{6}$ may have a solution is :

(a) $\left[\frac{1}{4}, \frac{1}{2}\right]$ (b) $[1, 3]$ (c) $\left[\frac{1}{6}, \frac{1}{2}\right]$ (d) $[2, 4]$

15. A real valued function $f(x)$ satisfies the functional equation $f(x - y) = f(x)f(y) - f(a - x)f(a + y)$ where 'a' is a given constant and $f(0) = 1$, $f(2a - x)$ is equal to :

(a) $-f(x)$ (b) $f(x)$ (c) $f(a) + f(a - x)$ (d) $f(-x)$

16. Let $g : R \to R$ be given by $g(x) = 3 + 4x$ if $g^n(x) = gogogo\ldots\ldots og(x)$ n times. Then inverse of $g^n(x)$ is equal to :

(a) $(x + 1 - 4^n)\cdot 4^{-n}$ (b) $(x - 1 + 4^n)4^{-n}$ (c) $(x + 1 + 4^n)4^{-n}$ (d) None of these

17. Let $f : D \to R$ be defined as : $f(x) = \frac{x^2 + 2x + a}{x^2 + 4x + 3a}$ where D and R denote the domain of f and the set of all real numbers respectively. If f is surjective mapping, then the complete range of a is :

(a) $0 \leq a \leq 1$ (b) $0 < a \leq 1$ (c) $0 \leq a < 1$ (d) $0 < a < 1$

18. If $f:(-\infty, 2] \longrightarrow (-\infty, 4]$, where $f(x) = x(4-x)$, then $f^{-1}(x)$ is given by :

(a) $2-\sqrt{4-x}$ (b) $2+\sqrt{4-x}$ (c) $-2+\sqrt{4-x}$ (d) $-2-\sqrt{4-x}$

19. If $[5\sin x] + [\cos x] + 6 = 0$, then range of $f(x) = \sqrt{3}\cos x + \sin x$ corresponding to solution set of the given equation is : (where $[\cdot]$ denotes greatest integer function)

(a) $[-2,-1)$ (b) $\left(-\frac{3\sqrt{3}+2}{5}, -1\right)$ (c) $[-2,-\sqrt{3})$ (d) $\left(-\frac{3\sqrt{3}+4}{5}, -1\right)$

20. If $f:R \to R$, $f(x) = ax + \cos x$ is an invertible function, then complete set of values of a is :

(a) $(-2,-1] \cup [1,2)$ (b) $[-1,1]$ (c) $(-\infty,-1] \cup [1,\infty)$ (d) $(-\infty,-2] \cup [2,\infty)$

21. The range of function $f(x) = [1+\sin x] + \left[2+\sin\frac{x}{2}\right] + \left[3+\sin\frac{x}{3}\right] + \ldots + \left[n+\sin\frac{x}{n}\right] \forall x \in [0,\pi]$, $n \in N$ ($[\cdot]$ denotes greatest integer function) is :

(a) $\left\{\frac{n^2+n-2}{2}, \frac{n(n+1)}{2}\right\}$ (b) $\left\{\frac{n(n+1)}{2}\right\}$

(c) $\left\{\frac{n(n+1)}{2}, \frac{n^2+n+2}{2}, \frac{n^2+n+4}{2}\right\}$ (d) $\left\{\frac{n(n+1)}{2}, \frac{n^2+n+2}{2}\right\}$

22. If $f:R \to R$, $f(x) = \frac{x^2+ax+1}{x^2+x+1}$, then the complete set of values of 'a' such that $f(x)$ is onto is :

(a) $(-\infty,\infty)$ (b) $(-\infty,0)$ (c) $(0,\infty)$ (d) not possible

23. If $f(x)$ and $g(x)$ are two functions such that $f(x) = [x] + [-x]$ and $g(x) = \{x\} \forall x \in R$ and $h(x) = f(g(x))$; then which of the following is incorrect ?

($[\cdot]$ denotes greatest integer function and $\{\cdot\}$ denotes fractional part function)

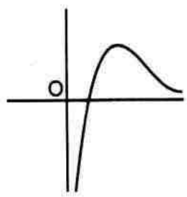
(a) $f(x)$ and $h(x)$ are identical functions (b) $f(x) = g(x)$ has no solution

(c) $f(x) + h(x) > 0$ has no solution (d) $f(x) - h(x)$ is a periodic function

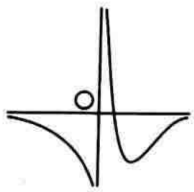
24. Number of elements in the range set of $f(x) = \left[\frac{x}{15}\right]\left[-\frac{15}{x}\right] \forall x \in (0,90)$; (where $[\cdot]$ denotes greatest integer function) :

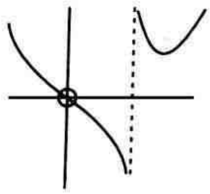
(a) 5 (b) 6 (c) 7 (d) Infinite

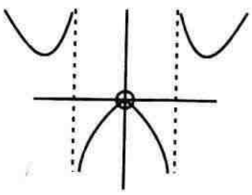
25. The graph of function $f(x)$ is shown below :

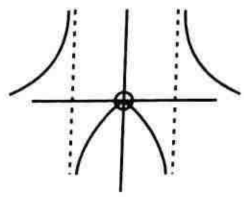


Then the graph of $g(x) = \frac{1}{f(|x|)}$ is :

(a) 

(b) 

(c) 

(d) 

26. Which of the following function is homogeneous ?

(a) $f(x) = x\sin y + y\sin x$

(b) $g(x) = xe^{\frac{y}{x}} + ye^{\frac{x}{y}}$

(c) $h(x) = \dfrac{xy}{x+y^2}$

(d) $\phi(x) = \dfrac{x - y\cos x}{y\sin x + y}$

27. Let $f(x) = \begin{bmatrix} 2x+3 & ; & x \le 1 \\ a^2x+1 & ; & x > 1 \end{bmatrix}$. If the range of $f(x) = R$ (set of real numbers) then number of integral value(s), which a may take :

(a) 2 (b) 3 (c) 4 (d) 5

28. The maximum integral value of x in the domain of $f(x) = \log_{10}(\log_{1/3}(\log_4(x-5)))$ is :

(a) 5 (b) 7 (c) 8 (d) 9

29. Range of the function $f(x) = \log_2\left(\dfrac{4}{\sqrt{x+2}+\sqrt{2-x}}\right)$ is :

(a) $(0, \infty)$ (b) $\left[\dfrac{1}{2}, 1\right]$ (c) $[1, 2]$ (d) $\left[\dfrac{1}{4}, 1\right]$

30. Number of integers statisfying the equation $|x^2+5x|+|x-x^2|=|6x|$ is :

(a) 3 (b) 5 (c) 7 (d) 9

31. Which of the following is not an odd function ?

(a) $\ln\left(\dfrac{x^4+x^2+1}{(x^2+x+1)^2}\right)$

(b) $\text{sgn}(\text{sgn}(x))$

(c) $\sin(\tan x)$

(d) $f(x)$, where $f(x) + f\left(\dfrac{1}{x}\right) = f(x)\cdot f\left(\dfrac{1}{x}\right)\ \forall\, x \in R - \{0\}$ and $f(2) = 33$

32. Which of the following function is periodic with fundamental period π?

(a) $f(x) = \cos x + \left[\left[\dfrac{\sin x}{2}\right]\right]$; where $[\cdot]$ denotes greatest integer function

(b) $g(x) = \dfrac{\sin x + \sin 7x}{\cos x + \cos 7x} + |\sin x|$

(c) $h(x) = \{x\} + |\cos x|$; where $\{\cdot\}$ denotes fractional part function

(d) $\phi(x) = |\cos x| + \ln(\sin x)$

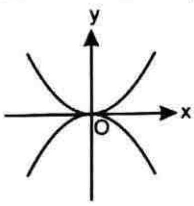
33. Let $f: N \longrightarrow Z$ and $f(x) = \begin{cases} \dfrac{x-1}{2} & ; \text{ when } x \text{ is odd} \\ -\dfrac{x}{2} & ; \text{ when } x \text{ is even} \end{cases}$, then :

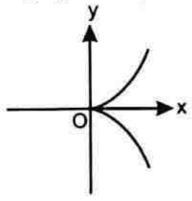
(a) $f(x)$ is bijective
(b) $f(x)$ is injective but not surjective
(c) $f(x)$ is not injective but surjective
(d) $f(x)$ is neither injective nor surjective

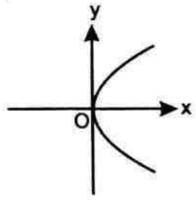
34. Let $g(x)$ be the inverse of $f(x) = \dfrac{2^{x+1} - 2^{1-x}}{2^x + 2^{-x}}$ then $g(x)$ be :

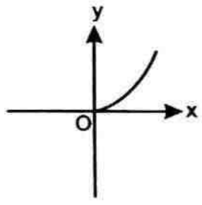
(a) $\dfrac{1}{2}\log_2\left(\dfrac{2+x}{2-x}\right)$ (b) $-\dfrac{1}{2}\log_2\left(\dfrac{2+x}{2-x}\right)$ (c) $\log_2\left(\dfrac{2+x}{2-x}\right)$ (d) $\log_2\left(\dfrac{2-x}{2+x}\right)$

35. Which of the following is the graph of the curve $\sqrt{|y|} = x$ is ?

(a) 

(b) 

(c) 

(d) 

36. Range of $f(x) = \log_{[x]}(9 - x^2)$; where $[\cdot]$ denotes G.I.F. is :

(a) $\{1, 2\}$ (b) $(-\infty, 2)$ (c) $(-\infty, \log_2 5]$ (d) $[\log_2 5, 3]$

37. If $e^x + e^{f(x)} = e$, then for $f(x)$:

(a) Domain is $(-\infty, 1)$ (b) Range is $(-\infty, 1]$ (c) Domain is $(-\infty, 0]$ (d) Range is $(-\infty, 0]$

38. If high voltage current is applied on the field given by the graph $y + |y| - x - |x| = 0$. On which of the following curve a person can move so that he remains safe ?

(a) $y = x^2$ (b) $y = \text{sgn}(-e^2)$ (c) $y = \log_{1/3} x$ (d) $y = m + |x|; m > 3$

39. If $|f(x) + 6 - x^2] = |f(x)| + |4 - x^2| + 2$, then $f(x)$ is necessarily non-negaive for :

(a) $x \in [-2, 2]$
(b) $x \in (-\infty, -2) \cup (2, \infty)$
(c) $x \in [-\sqrt{6}, \sqrt{6}]$
(d) $x \in [-5, -2] \cup [2, 5]$

40. Let $f(x) = \cos(px) + \sin x$ be periodic, then p must be :

(a) Positive real number
(b) Negative real number
(c) Rational
(d) Prime

41. The domain of $f(x)$ is (0, 1), therefore, the domain of $y = f(e^x) + f(\ln|x|)$ is :

(a) $\left(\frac{1}{e}, 1\right)$ (b) $(-e, -1)$ (c) $\left(-1, -\frac{1}{e}\right)$ (d) $(-e, -1) \cup (1, e)$

42. Let $A = \{1, 2, 3, 4\}$ and $f: A \to A$ satisfy $f(1) = 2, f(2) = 3, f(3) = 4, f(4) = 1$.
Suppose $g: A \to A$ satisfies $g(1) = 3$ and $fog = gof$, then g =

(a) $\{(1, 3), (2, 1), (3, 2), (4, 4)\}$ (b) $\{(1, 3), (2, 4), (3, 1), (4, 2)\}$
(c) $\{(1, 3), (2, 2), (3, 4), (4, 3)\}$ (d) $\{(1, 3), (2, 4), (3, 2), (4, 1)\}$

43. The number of solutions of the equation $[y + [y]] = 2\cos x$ is :
(where $y = \frac{1}{3}[\sin x + [\sin x + [\sin x]]]$ and $[\cdot]$ = greatest integer function)

(a) 0 (b) 1 (c) 2 (d) Infinite

44. The function, $f(x) = \begin{cases} \dfrac{(x^{2n})}{(x^{2n}\operatorname{sgn} x)^{2n+1}}\left(\dfrac{e^{\frac{1}{x}} - e^{-\frac{1}{x}}}{e^{\frac{1}{x}} + e^{-\frac{1}{x}}}\right) & x \neq 0 \\ 1 & x = 0 \end{cases}$ $n \in N$ is :

(a) Odd function (b) Even function
(c) Neither odd nor even function (d) Constant function

45. Let $f(1) = 1$, and $f(n) = 2\sum_{r=1}^{n-1} f(r)$. Then $\sum_{r=1}^{m} f(r)$ is equal to :

(a) $\frac{3^m - 1}{2}$ (b) 3^m (c) 3^{m-1} (d) $\frac{3^{m-1} - 1}{2}$

46. Let $f(x) = \frac{x}{\sqrt{1+x^2}}$, then $\underbrace{fofofo\ldots\ldots of}_{n\text{times}}(x)$ is :

(a) $\dfrac{x}{\sqrt{1+\left(\sum_{r=1}^{n} r\right)x^2}}$ (b) $\dfrac{x}{\sqrt{1+\left(\sum_{r=1}^{n} 1\right)x^2}}$ (c) $\left(\dfrac{x}{\sqrt{1+x^2}}\right)^n$ (d) $\dfrac{nx}{\sqrt{1+nx^2}}$

47. Let $f: R \to R$, $f(x) = 2x + |\cos x|$, then f is :

(a) One-one and into (b) One-one and onto
(c) Many-one and into (d) Many-one and onto

48. Let $f: R \to R$, $f(x) = x^3 + x^2 + 3x + \sin x$, then f is :

(a) One-one and into (b) One-one and onto
(c) Many-one and into (d) Many-one and onto

49. $f(x) = \{x\} + \{x+1\} + \{x+2\} + \ldots\ldots + \{x+99\}$, then $[f(\sqrt{2})]$, (where $\{\cdot\}$ denotes fractional part function and $[\cdot]$ denotes the greatest integer function) is equal to :

(a) 5050 (b) 4950 (c) 41 (d) 14

50. If $|\cot x + \operatorname{cosec} x| = |\cot x| + |\operatorname{cosec} x|$; $x \in [0, 2\pi]$, then complete set of values of x is :

(a) $[0, \pi]$ (b) $\left(0, \frac{\pi}{2}\right]$

(c) $\left(0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right)$ (d) $\left(\pi, \frac{3\pi}{2}\right] \cup \left[\frac{7\pi}{4}, 2\pi\right]$

51. The function $f(x) = 0$ has eight distinct real solution and f also satisfy $f(4+x) = f(4-x)$. The sum of all the eight solution of $f(x) = 0$ is :

(a) 12 (b) 32 (c) 16 (d) 15

52. Let $f(x)$ be a polynomial of degree 5 with leading coefficient unity such that $f(1) = 5, f(2) = 4, f(3) = 3, f(4) = 2, f(5) = 1$. Then $f(6)$ is equal to :

(a) 0 (b) 24 (c) 120 (d) 720

53. Let $f: A \to B$ be a function such that $f(x) = \sqrt{x-2} + \sqrt{4-x}$, is invertible, then which of the following is not possible ?

(a) $A = [3, 4]$ (b) $A = [2, 3]$ (c) $A = [2, 2\sqrt{3}]$ (d) $[2, 2\sqrt{2}]$

54. The number of positive integral values of x satisfying $\left[\frac{x}{9}\right] = \left[\frac{x}{11}\right]$ is :

(where [·] denotes greatest integer function)

(a) 21 (b) 22 (c) 23 (d) 24

55. The domain of function $f(x) = \log_{\left[x+\frac{1}{2}\right]}(2x^2 + x - 1)$, where [·] denotes the greatest integer function is :

(a) $\left[\frac{3}{2}, \infty\right)$ (b) $(2, \infty)$ (c) $\left(-\frac{1}{2}, \infty\right) - \left\{\frac{1}{2}\right\}$ (d) $\left(\frac{1}{2}, 1\right) \cup (1, \infty)$

56. The solution set of the equation $[x]^2 + [x+1] - 3 = 0$, where [·] represents greatest integer function is :

(a) $[-1, 0) \cup [1, 2)$ (b) $[-2, -1) \cup [1, 2)$ (c) $[1, 2)$ (d) $[-3, -2) \cup [2, 3)$

57. Which among the following relations is a function ?

(a) $x^2 + y^2 = r^2$ (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = r^2$ (c) $y^2 = 4ax$ (d) $x^2 = 4ay$

(where a, b, r are constants)

58. A funciton $f: R \to R$ is defined as $f(x) = 3x^2 + 1$. Then $f^{-1}(x)$ is :

(a) $\frac{\sqrt{x-1}}{3}$ (b) $\frac{1}{3}\sqrt{x} - 1$

(c) f^{-1} does not exist (d) $\sqrt{\frac{x-1}{3}}$

59. If $f(x) = \begin{cases} 2+x, & x \geq 0 \\ 4-x, & x < 0 \end{cases}$, then $f(f(x))$ is given by :

(a) $f(f(x)) = \begin{cases} 4+x &, x \geq 0 \\ 6-x &, x < 0 \end{cases}$ (b) $f(f(x)) = \begin{cases} 4+x &, x \geq 0 \\ x &, x < 0 \end{cases}$

(c) $f(f(x)) = \begin{cases} 4-x &, x \geq 0 \\ x &, x < 0 \end{cases}$ (d) $f(f(x)) = \begin{cases} 4-2x &, x \geq 0 \\ 4+2x &, x < 0 \end{cases}$

60. The function $f: R \to R$ defined as $f(x) = \dfrac{3x^2+3x-4}{3+3x-4x^2}$ is :

(a) One to one but not onto (b) Onto but not one to one

(c) Both one to one and onto (d) Neither one to one nor onto

61. The number of solutions of the equation $e^x - \log|x| = 0$ is :

(a) 0 (b) 1 (c) 2 (d) 3

62. If complete solution set of $e^{-x} \leq 4-x$ is $[\alpha, \beta]$, then $[\alpha]+[\beta]$ is equal to :

(where $[\cdot]$ denotes greatest integer function)

(a) 0 (b) 2 (c) 1 (d) 4

63. Range of $f(x) = \sqrt{\sin(\log_7(\cos(\sin x)))}$ is :

(a) $[0, 1)$ (b) $\{0, 1\}$ (c) $\{0\}$ (d) $[1, 7]$

64. If domain of $y = f(x)$ is $x \in [-3, 2]$, then domain of $y = f(|[x]|)$:

(where $[\cdot]$ denotes greatest integer function)

(a) $[-3, 2]$ (b) $[-2, 3)$ (c) $[-3, 3]$ (d) $[-2, 3]$

65. Range of the function $f(x) = \cot^{-1}\{-x\} + \sin^{-1}\{x\} + \cos^{-1}\{x\}$, where $\{\cdot\}$ denotes fractional part function :

(a) $\left(\dfrac{3\pi}{4}, \pi\right)$ (b) $\left[\dfrac{3\pi}{4}, \pi\right)$ (c) $\left[\dfrac{3\pi}{4}, \pi\right]$ (d) $\left(\dfrac{3\pi}{4}, \pi\right]$

66. Let $f: R - \left\{\dfrac{3}{2}\right\} \to R$, $f(x) = \dfrac{3x+5}{2x-3}$. Let $f_1(x) = f(x)$, $f_n(x) = f(f_{n-1}(x))$ for $n \geq 2$, $n \in N$, then $f_{2008}(x) + f_{2009}(x) =$

(a) $\dfrac{2x^2+5}{2x-3}$ (b) $\dfrac{x^2+5}{2x-3}$ (c) $\dfrac{2x^2-5}{2x-3}$ (d) $\dfrac{x^2-5}{2x-3}$

67. Range of the function, $f(x) = \dfrac{(1+x+x^2)(1+x^4)}{x^3}$, for $x > 0$ is :

(a) $[0, \infty)$ (b) $[2, \infty)$ (c) $[4, \infty)$ (d) $[6, \infty)$

68. The function $f: (-\infty, 3] \to (0, e^7]$ defined by $f(x) = e^{x^3-3x^2-9x+2}$ is :

(a) Many-one and onto (b) Many-one and into

(c) One to one and onto (d) One to one and into

69. If $f(x) = \sin\left\{\log\left(\frac{\sqrt{4-x^2}}{1-x}\right)\right\}$; $x \in R$, then range of $f(x)$ is given by :

(a) $[-1, 1]$ (b) $[0, 1]$ (c) $(-1, 1)$ (d) None of these

70. Set of values of 'a' for which the function $f: R \to R$, given by $f(x) = x^3 + (a+2)x^2 + 3ax + 10$ is one-one is given by :

(a) $(-\infty, 1] \cup [4, \infty)$ (b) $[1, 4]$ (c) $[1, \infty)$ (d) $[-\infty, 4]$

71. If the range of the function $f(x) = \tan^{-1}(3x^2 + bx + c)$ is $\left[0, \frac{\pi}{2}\right)$; (domain is R), then :

(a) $b^2 = 3c$ (b) $b^2 = 4c$ (c) $b^2 = 12c$ (d) $b^2 = 8c$

72. Let $f(x) = \sin^{-1} x - \cos^{-1} x$, then the set of values of k for which of $|f(x)| = k$ has exactly two distinct solutions is :

(a) $\left(0, \frac{\pi}{2}\right]$ (b) $\left(0, \frac{\pi}{2}\right)$ (c) $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (d) $\left[\pi, \frac{3\pi}{2}\right]$

73. Let $f: R \to R$ is defined by $f(x) = \begin{cases} (x+1)^3 & ; \ x \le 1 \\ \ln x + (b^2 - 3b + 10) & ; \ x > 1 \end{cases}$. If $f(x)$ is invertible, then the set of all values of 'b' is :

(a) $\{1, 2\}$ (b) ϕ (c) $\{2, 5\}$ (d) None of these

74. Let $f(x)$ is continuous function with range $[-1, 1]$ and $f(x)$ is defined $\forall\ x \in R$. If $g(x) = \frac{e^{f(x)} - e^{|f(x)|}}{e^{f(x)} + e^{|f(x)|}}$, then range of $g(x)$ is :

(a) $[0, 1]$ (b) $\left[0, \frac{e^2+1}{e^2-1}\right]$

(c) $\left[0, \frac{e^2-1}{e^2+1}\right]$ (d) $\left[\frac{-e^2+1}{e^2+1}, 0\right]$

75. Consider all functions $f: \{1, 2, 3, 4\} \to \{1, 2, 3, 4\}$ which are one-one, onto and satisfy the following property :

if $f(k)$ is odd then $f(k+1)$ is even, $k = 1, 2, 3$.

The number of such functions is :

(a) 4 (b) 8 (c) 12 (d) 16

76. Consider the function $f: R - \{1\} \to R - \{2\}$ given by $f(x) = \frac{2x}{x-1}$. Then :

(a) f is one-one but not onto (b) f is onto but not one-one

(c) f is neither one-one nor onto (d) f is both one-one and onto

77. If range of function $f(x)$ whose domain is set of all real numbers is $[-2, 4]$, then range of function $g(x) = \frac{1}{2} f(2x+1)$ is equal to :

(a) $[-2, 4]$ (b) $[-1, 2]$ (c) $[-3, 9]$ (d) $[-2, 2]$

78. Let $f : R \to R$ and $f(x) = \frac{x(x^4+1)(x+1)+x^4+2}{x^2+x+1}$, then $f(x)$ is :

(a) One-one, into (b) Many-one, onto
(c) One-one, onto (d) Many one, into

79. Let $f(x)$ be defined as :

$$f(x) = \begin{cases} |x| & 0 \le x < 1 \\ |x-1| + |x-2| & 1 \le x < 2 \\ |x-3| & 2 \le x < 3 \end{cases}$$

The range of function $g(x) = \sin(7(f(x))$ is :

(a) $[0, 1]$ (b) $[-1, 0]$ (c) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (d) $[-1, 1]$

80. If $[x]^2 - 7[x] + 10 < 0$ and $4[y]^2 - 16[y] + 7 < 0$, then $[x+y]$ cannot be ([·] denotes greatest integer function) :

(a) 7 (b) 8 (c) 9 (d) both (b) and (c)

81. Let $f : R \to R$ be a function defined by $f(x) = \frac{e^{|x|} - e^{-x}}{e^x + e^{-x}}$, then

(a) $f(x)$ is many one, onto function (b) $f(x)$ is many one, into function
(c) $f(x)$ is decreasing function $\forall x \in R$ (d) $f(x)$ is bijective function

82. The function $f(x)$ satisfy the equation $f(1-x) + 2f(x) = 3x \ \forall x \in R$, then $f(0) =$

(a) -2 (b) -1 (c) 0 (d) 1

83. Let $f : [0, 5] \to [0, 5]$ be an invertible function defined by $f(x) = ax^2 + bx + c$, where $a, b, c \in R$, $abc \ne 0$, then one of the root of the equation $cx^2 + bx + a = 0$ is :

(a) a (b) b (c) c (d) $a+b+c$

84. Let $f(x) = x^2 + \lambda x + \mu \cos x$, λ being an integer and μ is a real number. The number of ordered pairs (λ, μ) for which the equation $f(x) = 0$ and $f(f(x)) = 0$ have the same (non empty) set of real roots is :

(a) 2 (b) 3 (c) 4 (d) 6

85. Consider all function $f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4\}$ which are one-one, onto and satisfy the following property :

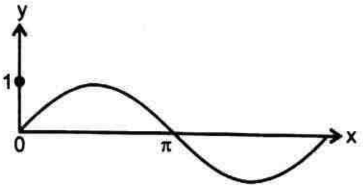
if $f(k)$ is odd then $f(k+1)$ is even, $k = 1, 2, 3$.

The number of such function is :

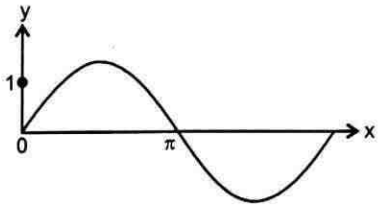
(a) 4 (b) 8 (c) 12 (d) 16

86. Which of the following is closest to the graph of $y = \tan(\sin x), x > 0$?

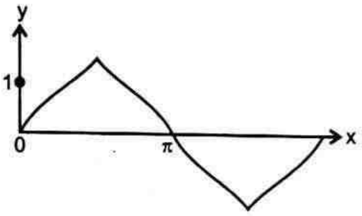
(a)



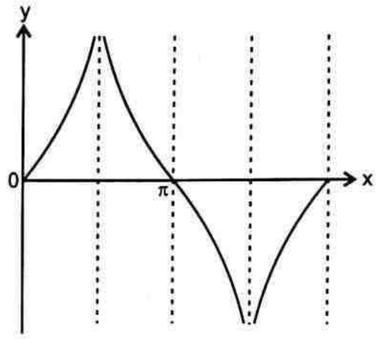
(b)



(c)



(d)



87. Consider the function $f : R - \{1\} \to R - \{2\}$ given by $f(x) = \dfrac{2x}{x-1}$. Then

(a) f is one-one but not onto
(b) f is onto but not one-one
(c) f is neither one-one nor onto
(d) f is both one-one and onto

88. If range of function $f(x)$ whose domain is set of all real numbers is $[-2, 4]$, then range of function $g(x) = \dfrac{1}{2} f(2x+1)$ is equal to :

(a) $[-2, 4]$
(b) $[-1, 2]$
(c) $[-3, 9]$
(d) $[-2, 2]$

89. Let $f : R \to R$ and $f(x) = \dfrac{x(x^4+1)(x+1)+x^4+2}{x^2+x+1}$, then $f(x)$ is :

(a) One-one, into
(b) Many one, onto
(c) One-one, onto
(d) Many one, into

90. Let $f(x)$ be defined as

$$f(x) = \begin{cases} |x| & 0 \le x < 1 \\ |x-1| + |x-2| & 1 \le x < 2 \\ |x-3| & 2 \le x < 3 \end{cases}$$

The range of function $g(x) = \sin(7(f(x))$ is :

(a) $[0, 1]$
(b) $[-1, 0]$
(c) $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$
(d) $[-1, 1]$

91. The number of integral values of x in the domain of function f defined as $f(x) = \sqrt{\ln|\ln|x||} + \sqrt{7|x| - |x|^2 - 10}$ is :

(a) 5
(b) 6
(c) 7
(d) 8

92. The complete set of values of x in the domain of function $f(x)=\sqrt{\log_{x+2\{x\}}([x]^2-5[x]+7)}$ (where [·] denote greatest integer function and {·} denote fraction part function) is :

(a) $\left(-\frac{1}{3},0\right)\cup\left(\frac{1}{3},1\right)\cup(2,\infty)$ (b) $(0,1)\cup(1,\infty)$

(c) $\left(-\frac{2}{3},0\right)\cup\left(\frac{1}{3},1\right)\cup(1,\infty)$ (d) $\left(-\frac{1}{3},0\right)\cup\left(\frac{1}{3},1\right)\cup(1,\infty)$

93. The number of integral ordered pair (x, y) that satisfy the system of equation $|x+y-4|=5$ and $|x-3|+|y-1|=5$ is/are :

(a) 2 (b) 4 (c) 6 (d) 12

94. Let $f:R\to R$, where $f(x)=\dfrac{x^2+ax+1}{x^2+x+1}$. Then the complete set of values of 'a' such that $f(x)$ is onto is :

(a) $(-\infty,\infty)$ (b) $(-\infty,0)$ (c) $(0,\infty)$ (d) Empty set

95. If $A=\{1,2,3,4\}$ and $f:A\to A$, then total number of invertible function 'f' such that $f(2)\neq 2$, $f(4)\neq 4$, $f(1)=1$ is equal to :

(a) 1 (b) 2 (c) 3 (d) 4

96. The domain of definition of $f(x)=\log_{(x^2-x+1)}(2x^2-7x+9)$ is :

(a) R (b) $R-\{0\}$ (c) $R-\{0,1\}$ (d) $R-\{1\}$

97. If $A=\{1,2,3,4\}, B=\{1,2,3,4,5,6\}$ and $f:A\to B$ is an injective mapping satisfying $f(i)\neq i$, then number of such mappings are :

(a) 182 (b) 181 (c) 183 (d) none of these

98. Let $f(x)=x^2-2x-3; x\geq 1$ and $g(x)=1+\sqrt{x+4}; x\geq -4$ then the number of real solutions of equation $f(x)=g(x)$ is/are

(a) 0 (b) 1 (c) 2 (d) 4

Answers

1.	(b)	**2.**	(c)	**3.**	(c)	**4.**	(b)	**5.**	(d)	**6.**	(a)	**7.**	(d)	**8.**	(c)	**9.**	(c)	**10.**	(a)
11.	(b)	**12.**	(c)	**13.**	(c)	**14.**	(b)	**15.**	(a)	**16.**	(a)	**17.**	(d)	**18.**	(a)	**19.**	(d)	**20.**	(c)
21.	(d)	**22.**	(d)	**23.**	(b)	**24.**	(b)	**25.**	(c)	**26.**	(b)	**27.**	(c)	**28.**	(c)	**29.**	(b)	**30.**	(c)
31.	(d)	**32.**	(b)	**33.**	(a)	**34.**	(c)	**35.**	(b)	**36.**	(c)	**37.**	(a)	**38.**	(d)	**39.**	(a)	**40.**	(c)
41.	(b)	**42.**	(b)	**43.**	(a)	**44.**	(b)	**45.**	(c)	**46.**	(b)	**47.**	(b)	**48.**	(b)	**49.**	(c)	**50.**	(c)
51.	(b)	**52.**	(c)	**53.**	(c)	**54.**	(d)	**55.**	(a)	**56.**	(b)	**57.**	(d)	**58.**	(c)	**59.**	(a)	**60.**	(b)
61.	(b)	**62.**	(c)	**63.**	(c)	**64.**	(b)	**65.**	(d)	**66.**	(a)	**67.**	(d)	**68.**	(a)	**69.**	(a)	**70.**	(b)
71.	(c)	**72.**	(a)	**73.**	(a)	**74.**	(d)	**75.**	(c)	**76.**	(d)	**77.**	(b)	**78.**	(d)	**79.**	(d)	**80.**	(c)
81.	(b)	**82.**	(b)	**83.**	(a)	**84.**	(c)	**85.**	(c)	**86.**	(b)	**87.**	(d)	**88.**	(b)	**89.**	(d)	**90.**	(d)
91.	(b)	**92.**	(d)	**93.**	(d)	**94.**	(d)	**95.**	(c)	**96.**	(c)	**97.**	(b)	**98.**	(b)				

Exercise-2 : One or More than One Answer is/are Correct

1. $f(x)$ is an even periodic function with period 10. In [0, 5], $f(x) = \begin{cases} 2x & 0 \le x < 2 \\ 3x^2 - 8 & 2 \le x < 4 \\ 10x & 4 \le x \le 5 \end{cases}$. Then :

(a) $f(-4) = 40$
(b) $\dfrac{f(-13) - f(11)}{f(13) + f(-11)} = \dfrac{17}{21}$
(c) $f(5)$ is not defined
(d) Range of $f(x)$ is $[0, 50]$

2. Let $f(x) = ||x^2 - 4x + 3| - 2|$. Which of the following is/are correct ?

(a) $f(x) = m$ has exactly two real solutions of different sign $\forall\, m > 2$
(b) $f(x) = m$ has exactly two real solutions $\forall\, m \in (2, \infty) \cup \{0\}$
(c) $f(x) = m$ has no solutions $\forall\, m < 0$
(d) $f(x) = m$ has four distinct real solution $\forall\, m \in (0, 1)$

3. Let $f(x) = \cos^{-1}\left(\dfrac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}\right)$

Which of the following statement(s) is/are correct about $f(x)$?

(a) Domain is R
(b) Range is $[0, \pi]$
(c) $f(x)$ is even
(d) $f(x)$ is derivable in $(\pi, 2\pi)$

4. $|\log_e |x|| = |k - 1| - 3$ has four distinct roots then k satisfies : (where $|x| < e^2, x \ne 0$)

(a) $(-4, -2)$
(b) $(4, 6)$
(c) (e^{-1}, e)
(d) (e^{-2}, e^{-1})

5. Which of the following functions are defined for all $x \in R$?
(Where $[\cdot]$ = denotes greatest integer function)

(a) $f(x) = \sin[x] + \cos[x]$
(b) $f(x) = \sec^{-1}(1 + \sin^2 x)$
(c) $f(x) = \sqrt{\dfrac{9}{8} + \cos x + \cos 2x}$
(d) $f(x) = \tan(\ln(1 + |x|))$

6. Let $f(x) = \begin{cases} x^2 & 0 < x < 2 \\ 2x - 3 & 2 \le x < 3 \\ x + 2 & x \ge 3 \end{cases}$, then the true equations :

(a) $f\left(f\left(f\left(\dfrac{3}{2}\right)\right)\right) = f\left(\dfrac{3}{2}\right)$
(b) $1 + f\left(f\left(f\left(\dfrac{5}{2}\right)\right)\right) = f\left(\dfrac{5}{2}\right)$
(c) $f(f(f(2))) = f(1)$
(d) $\underbrace{f(f(f(\ldots\ldots f(4))\ldots)}_{1004\text{ times}} = 2012$

7. Let $f : \left[\dfrac{2\pi}{3}, \dfrac{5\pi}{3}\right] \longrightarrow [0, 4]$ be a function defined as $f(x) = \sqrt{3}\sin x - \cos x + 2$, then :

(a) $f^{-1}(1) = \dfrac{4\pi}{3}$
(b) $f^{-1}(1) = \pi$
(c) $f^{-1}(2) = \dfrac{5\pi}{6}$
(d) $f^{-1}(2) = \dfrac{7\pi}{6}$

8. Let $f(x)$ be invertible function and let $f^{-1}(x)$ be its inverse. Let equation $f(f^{-1}(x)) = f^{-1}(x)$ has two real roots α and β (with in domain of $f(x)$), then :

(a) $f(x) = x$ also have same two real roots

(b) $f^{-1}(x) = x$ also have same two real roots

(c) $f(x) = f^{-1}(x)$ also have same two real roots

(d) Area of triangle formed by $(0, 0)$, $(\alpha, f(\alpha))$, and $(\beta, f(\beta))$ is 1 unit

9. The function $f(x) = \cos^{-1} x + \cos^{-1}\left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right)$, then :

(a) Range of $f(x)$ is $\left[\frac{\pi}{3}, \frac{10\pi}{3}\right]$

(b) Range of $f(x)$ is $\left[\frac{\pi}{3}, \frac{5\pi}{3}\right]$

(c) $f(x)$ is one-one for $x \in \left[-1, \frac{1}{2}\right]$

(d) $f(x)$ is one-one for $x \in \left[\frac{1}{2}, 1\right]$

10. Let $f: R \to R$ defined by $f(x) = \cos^{-1}(-\{-x\})$, where $\{x\}$ is fractional part function. Then which of the following is/are correct ?

(a) f is many-one but not even function

(b) Range of f contains two prime numbers

(c) f is a periodic

(d) Graph of f does not lie below x-axis

11. Which option(s) is/are true ?

(a) $f: R \to R$, $f(x) = e^{|x|} - e^{-x}$ is many-one into function

(b) $f: R \to R$, $f(x) = 2x + |\sin x|$ is one-one onto

(c) $f: R \to R$, $f(x) = \dfrac{x^2 + 4x + 30}{x^2 - 8x + 18}$ is many-one onto

(d) $f: R \to R$, $f(x) = \dfrac{2x^2 - x + 5}{7x^2 + 2x + 10}$ is many-one into

12. If $h(x) = \left[\ln \frac{x}{e}\right] + \left[\ln \frac{e}{x}\right]$, where $[\cdot]$ denotes greatest integer function, then which of the following are true ?

(a) range of $h(x)$ is $\{-1, 0\}$

(b) If $h(x) = 0$, then x must be irrational

(c) If $h(x) = -1$, then x can be rational as well as irrational

(d) $h(x)$ is periodic function

13. If $f(x) = \begin{cases} x^3 & ; \ x \in Q \\ -x^3 & ; \ x \notin Q \end{cases}$, then :

(a) $f(x)$ is periodic

(b) $f(x)$ is many-one

(c) $f(x)$ is one-one

(d) range of the function is R

14. Let $f(x)$ be a real valued continuous function such that

$f(0) = \frac{1}{2}$ and $f(x+y) = f(x)f(a-y) + f(y)f(a-x) \ \forall \ x, y \in R,$

then for some real a :

(a) $f(x)$ is a periodic function
(b) $f(x)$ is a constant function
(c) $f(x) = \frac{1}{2}$
(d) $f(x) = \frac{\cos x}{2}$

15. $f(x)$ is an even periodic function with period 10. In $[0, 5]$, $f(x) = \begin{cases} 2x & 0 \le x < 2 \\ 3x^2 - 8 & 2 \le x < 4 \\ 10x & 4 \le x \le 5 \end{cases}$. Then :

(a) $f(-4) = 40$
(b) $\frac{f(-13) - f(11)}{f(13) + f(-11)} = \frac{17}{21}$
(c) $f(5)$ is not defined
(d) Range of $f(x)$ is $[0, 50]$

16. For the equation $\frac{e^{-x}}{1+x} = \lambda$ which of the following statement(s) is/are correct ?

(a) when $\lambda \in (0, \infty)$ equation has 2 real and distinct roots
(b) when $\lambda \in (-\infty, -e^2)$ equation has 2 real and distinct roots
(c) when $\lambda \in (0, \infty)$ equation has 1 real root
(d) when $\lambda \in (-e, 0)$ equation has no real root

17. For $x \in R^+$, if $x, [x], \{x\}$ are in harmonic progression then the value of x can not be equal to :
(where $[\cdot]$ denotes greatest integer function, $\{\cdot\}$ denotes fractional part function)

(a) $\frac{1}{\sqrt{2}} \tan \frac{\pi}{8}$
(b) $\frac{1}{\sqrt{2}} \cot \frac{\pi}{8}$
(c) $\frac{1}{\sqrt{2}} \tan \frac{\pi}{12}$
(d) $\frac{1}{\sqrt{2}} \cot \frac{\pi}{12}$

18. The equation $||x-1|+a| = 4$, $a \in R$, has :

(a) 3 distinct real roots for unique value of a.
(b) 4 distinct real roots for $a \in (-\infty, -4)$
(c) 2 distinct real roots for $|a| < 4$
(d) no real roots for $a > 4$

19. Let $f_n(x) = (\sin x)^{1/n} + (\cos x)^{1/n}$, $x \in R$, then :

(a) $f_2(x) > 1$ for all $x \in \left(2k\pi, (4k+1)\frac{\pi}{2}\right)$, $k \in I$
(b) $f_2(x) = 1$ for $x = 2k\pi$, $k \in I$
(c) $f_2(x) > f_3(x)$ for all $x \in \left(2k\pi, (4k+1)\frac{\pi}{2}\right)$, $k \in I$
(d) $f_3(x) \ge f_5(x)$ for all $x \in \left(2k\pi, (4k+1)\frac{\pi}{2}\right)$, $k \in I$

(Where I denotes set of integers)

20. If the domain of $f(x) = \frac{1}{\pi}\cos^{-1}\left[\log_3\left(\frac{x^2}{3}\right)\right]$ where, $x > 0$ is $[a, b]$ and the range of $f(x)$ is $[c, d]$, then :

(a) a, b are the roots of the equation $x^4 - 3x^3 - x + 3 = 0$

(b) a, b are the roots of the equation $x^4 - x^3 + x^2 - 2x + 1 = 0$

(c) $a^3 + d^3 = 1$

(d) $a^2 + b^2 + c^2 + d^2 = 11$

21. The number of real values of x satisfying the equation ; $\left[\frac{2x+1}{3}\right] + \left[\frac{4x+5}{6}\right] = \frac{3x-1}{2}$ are greater than or equal to {[·] denotes greatest integer function):

(a) 7 (b) 8 (c) 9 (d) 10

22. Let $f(x) = \sin^6\left(\frac{x}{4}\right) + \cos^6\left(\frac{x}{4}\right)$. If $f^n(x)$ denotes n^{th} derivative of f evaluated at x. Then which of the following hold ?

(a) $f^{2014}(0) = -\frac{3}{8}$ (b) $f^{2015}(0) = \frac{3}{8}$ (c) $f^{2010}\left(\frac{\pi}{2}\right) = 0$ (d) $f^{2011}\left(\frac{\pi}{2}\right) = \frac{3}{8}$

23. Which of the following is(are) incorrect ?

(a) If $f(x) = \sin x$ and $g(x) = \ln x$ then range of $g(f(x))$ is $[-1, 1]$

(b) If $x^2 + ax + 9 > x \, \forall \, x \in R$ then $-5 < a < 7$

(c) If $f(x) = (2011 - x^{2012})^{\frac{1}{2012}}$ then $f(f(2)) = \frac{1}{2}$

(d) The function $f : R \to R$ defined as $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$ is not surjective.

24. If $[x]$ denotes the integral part of x for real x, and

$$S = \left[\frac{1}{4}\right] + \left[\frac{1}{4} + \frac{1}{200}\right] + \left[\frac{1}{4} + \frac{1}{100}\right] + \left[\frac{1}{4} + \frac{3}{200}\right] \ldots\ldots + \left[\frac{1}{4} + \frac{199}{200}\right] \text{ then}$$

(a) S is a composite number

(b) Exponent of S in $\lfloor 100$ is 12

(c) Number of factors of S is 10

(d) ${}^{25}C_r$ is max when $r = 51$

Answers

1.	(a, b, d)	**2.**	(a, b, c)	**3.**	(c, d)	**4.**	(a, b)	**5.**	(a, b, c)	**6.**	(a, b, c, d)
7.	(a, d)	**8.**	(a, b, c)	**9.**	(b, c)	**10.**	(a, b, d)	**11.**	(a, b, d)	**12.**	(a, c)
13.	(c, d)	**14.**	(a, b, c)	**15.**	(a, b, d)	**16.**	(b, c, d)	**17.**	(a, c, d)	**18.**	(a, b, c, d)
19.	(a, b)	**20.**	(a, d)	**21.**	(a, b, c)	**22.**	(a, c, d)	**23.**	a, b)	**24.**	(a, b)

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 3

Let $f(x) = \log_{\{x\}}[x]$

$g(x) = \log_{\{x\}}\{x\}$

$h(x) \log_{[x]}\{x\}$

where [ ], { } denotes the greatest integer function and fractional part fucntion respectively.

1. For $x \in (1, 5)$ the $f(x)$ is not defined at how many points :

(a) 5 (b) 4 (c) 3 (d) 2

2. If $A = \{x : x \in$ domain of $f(x)\}$ and $B = \{x : x \in$ domain of $g(x)\}$ then $\forall\ x \in (1, 5)$, $A - B$ will be :

(a) $(2, 3)$ (b) $(1, 3)$ (c) $(1, 2)$ (d) None of these

3. Domain of $h(x)$ is :

(a) $[2, \infty)$ (b) $[1, \infty)$ (c) $[2, \infty) - \{I\}$ (d) $R^+ - \{I\}$

I denotes integers.

Paragraph for Question Nos. 4 to 6

θ is said to be well behaved if it lies in interval $\left[0, \frac{\pi}{2}\right]$. They are intelligent if they make domain of $f + g$ and g equal. The values of θ for which $h(\theta)$ is defined are handsome. Let

$$f(x) = \sqrt{\theta x^2 - 2(\theta^2 - 3)x - 12\theta}, g(x) = \ln(x^2 - 49),$$

$$h(\theta) = \ln\left[\int_0^\theta 4\cos^2 t\, dt - \theta^2\right], \text{ where } \theta \text{ is in radians.}$$

4. Complete set of values of θ which are well behaved as well as intelligent is :

(a) $\left[\frac{3}{4}, \frac{\pi}{2}\right]$ (b) $\left[\frac{3}{5}, \frac{7}{8}\right]$ (c) $\left[\frac{5}{6}, \frac{\pi}{2}\right]$ (d) $\left[\frac{6}{7}, \frac{\pi}{2}\right]$

5. Complete set of values of θ which are intelligent is :

(a) $\left[\frac{6}{7}, \frac{7}{2}\right]$ (b) $\left(0, \frac{\pi}{3}\right]$ (c) $\left[\frac{1}{4}, \frac{6}{7}\right]$ (d) $\left[\frac{1}{2}, \frac{\pi}{2}\right]$

6. Complete set of values of θ which are well behaved, intelligent and handsome is :

(a) $\left(0, \frac{\pi}{2}\right]$ (b) $\left[\frac{6}{7}, \frac{\pi}{2}\right]$ (c) $\left[\frac{3}{4}, \frac{\pi}{2}\right]$ (d) $\left[\frac{3}{5}, \frac{\pi}{2}\right]$

Paragraph for Question Nos. 7 to 8

Let $f(x) = 2 - |x-3|, 1 \le x \le 5$ and for rest of the values $f(x)$ can be obtained by using the relation $f(5x) = \alpha f(x) \ \forall \ x \in R$.

7. The maximum value of $f(x)$ in $[5^4, 5^5]$ for $\alpha = 2$ is :

(a) 16 (b) 32 (c) 64 (d) 8

8. The value of $f(2007)$, taking $\alpha = 5$, is :

(a) 1118 (b) 2007 (c) 1250 (d) 132

Paragraph for Question Nos. 9 to 10

An even periodic function $f : R \to R$ with period 4 is such that

$$f(x) = \begin{cases} \max.(|x|, x^2) & ; \ 0 \le x < 1 \\ x & ; \ 1 \le x \le 2 \end{cases}$$

9. The value of $\{f(5.12)\}$ (where $\{\cdot\}$ denotes fractional part function), is :

(a) $\{f(3.26)\}$ (b) $\{f(7.88)\}$ (c) $\{f(2.12)\}$ (d) $\{f(5.88)\}$

10. The number of solutions of $f(x) = |3 \sin x|$ for $x \in (-6, 6)$ are :

(a) 5 (b) 3 (c) 7 (d) 9

Paragraph for Question Nos. 11 to 12

Let $f(x) = \dfrac{2|x|-1}{x-3}$

11. Range of $f(x)$:

(a) $R - \{3\}$ (b) $\left(-\infty, \dfrac{1}{3}\right] \cup (2, \infty)$ (c) $\left(-2, \dfrac{1}{3}\right] \cup (2, \infty)$ (d) R

12. Range of the values of 'k' for which $f(x) = k$ has exactly two distinct solutions :

(a) $\left(-2, \dfrac{1}{3}\right)$ (b) $(-2, 1]$ (c) $\left(0, \dfrac{2}{3}\right]$ (d) $(-\infty, -2)$

Paragraph for Question Nos. 13 to 14

Let $f(x)$ be a continuous function (define for all x) which satisfies $f^3(x) - 5f^2(x) + 10f(x) - 12 \ge 0$, $f^2(x) - 4f(x) + 3 \ge 0$ and $f^2(x) - 5f(x) + 6 \le 0$

13. If distinct positive number b_1, b_2 and b_3 are in G.P. then $f(1) + \ln b_1$, $f(2) + \ln b_2$, $f(3) + \ln b_3$ are in :

(a) A.P. (b) G.P. (c) H.P. (d) A.G.P.

14. The equation of tangent that can be drawn from (2, 0) on the curve $y = x^2 f(\sin x)$ is :

(a) $y = 24(x+2)$ (b) $y = 12(x+2)$ (c) $y = 24(x-2)$ (d) $y = 12(x-2)$

Paragraph for Question Nos. 15 to 16

Let $f:[2,\infty)\to[1,\infty)$ defined by $f(x)=2^{x^4-4x^2}$ and $g:\left[\frac{\pi}{2},\pi\right]\to A$ defined by $g(x)=\dfrac{\sin x+4}{\sin x-2}$ be two invertible functions, then

15. $f^{-1}(x)$ is equal to

(a) $\sqrt{2+\sqrt{4-\log_2 x}}$ (b) $\sqrt{2+\sqrt{4+\log_2 x}}$ (c) $\sqrt{4+\sqrt{2+\log_2 x}}$ (d) $\sqrt{4-\sqrt{2+\log_2 x}}$

16. The set 'A' equals to

(a) $[5,2]$ (b) $[-2,5]$ (c) $[-5,2]$ (d) $[-5,-2]$

Answers

1.	(c)	**2.**	(d)	**3.**	(c)	**4.**	(d)	**5.**	(a)	**6.**	(b)	**7.**	(b)	**8.**	(a)	**9.**	(b)	**10.**	(c)
11.	(b)	**12.**	(a)	**13.**	(a)	**14.**	(c)	**15.**	(b)	**16.**	(d)								

Exercise-4 : Matching Type Problems

1. If $x, y, z \in R$ satisfies the system of equations $x + [y] + \{z\} = 12.7$, $[x] + \{y\} + z = 4.1$ and $\{x\} + y + [z] = 2$
(where $\{\cdot\}$ and $[\cdot]$ denotes the fractional and integral parts respectively), then match the following :

	Column-I		Column-II
(A)	$\{x\} + \{y\} =$	**(P)**	7.7
(B)	$[z] + [x] =$	**(Q)**	1.1
(C)	$x + \{z\} =$	**(R)**	1
(D)	$z + [y] - \{x\} =$	**(S)**	3
		(T)	4

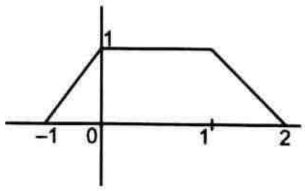
2. Consider $ax^4 + (7a - 2b)x^3 + (12a - 14b - c)x^2 - (24b + 7c)x + 1 - 12c = 0$, has no real roots

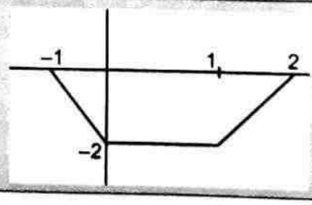
and $f_1(x) = \dfrac{\sqrt{\log_{(\pi+e)}(ax^4 + (7a-2b)x^3 + (12a - 14b - c)x^2 - (24b + 7c)x + 1 - 12c)}}{\sqrt{a}\sqrt{-\text{sgn}(1 + ac + b^2)}}$

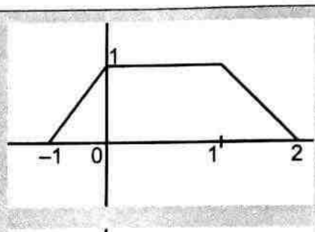
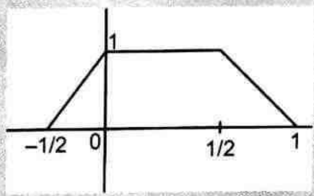
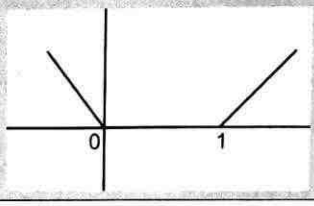
$f_2(x) = -2 + 2\log_{\sqrt{2}} \cos\left(\tan^{-1}(\sin(\pi(\cos(\pi(x + \frac{7}{2})))))\right)$. Then match the following :

	Column-I		Column-II
(A)	Domain of $f_1(x)$ is	**(P)**	$[-3, -2]$
(B)	Range of $f_2(x)$ in the domain of $f_1(x)$ is	**(Q)**	$[-4, -2]$
(C)	Range of $f_2(x)$ is	**(R)**	$(-\infty, \infty)$
(D)	Domain of $f_2(x)$ is	**(S)**	$(-\infty, -4] \cup [-3, \infty)$
		(T)	$[0, 1]$

3. Given the graph of $y = f(x)$



	Column-I		Column-II
(A)	$y = f(1 - x)$	**(P)**	

(B)	$y = f(2x)$	**(Q)**	
(C)	$y = -2f(x)$	**(R)**	
(D)	$y = 1 - f(x)$	**(S)**	

4.

	Column-I		Column-II
(A)	$f(x) = \sin^2 2x - 2\sin^2 x$	**(P)**	Range contains no natural number
(B)	$f(x) = \frac{4}{\pi}(\sin^{-1}(\sin \pi x))$	**(Q)**	Range contains atleast one integer
(C)	$f(x) = \sqrt{\ln(\cos(\sin x)}$	**(R)**	Many one but not even function
(D)	$f(x) = \tan^{-1}\left(\frac{x^2+1}{x^2+\sqrt{3}}\right)$	**(S)**	Both many one and even function
		(T)	Periodic but not odd function

5.

	Column-I		Column-II
(A)	If $\|x^2 - x\| \ge x^2 + x$, then complete set of values of x is	**(P)**	$(0, \infty)$
(B)	If $\|x + y\| > x - y$, where $x > 0$, then complete set of values of y is	**(Q)**	$(-\infty, 0]$
(C)	If $\log_2 x \ge \log_2(x^2)$, then complete set of values of x is	**(R)**	$[-1, \infty)$

(D)	$[x]+2 \geq |x|$, (where $[\cdot]$ denotes the greatest integer function) then complete set of values of x is	**(S)**	$(0,1]$
		(T)	$[1,\infty)$

6.

	Column-I		Column-II
(A)	Domain of $f(x)=\ln\tan^{-1}\{(x^3-6x^2+11x-6)\,x(e^x-1)\}$ is	**(P)**	$\left[-1,\frac{5}{4}\right]$
(B)	Range of $f(x)=\sin^2\frac{x}{4}+\cos\frac{x}{4}$ is	**(Q)**	$[2,\infty)$
(C)	The domain of function $f(x)=\sqrt{\log_{(\lvert x\rvert-1)}(x^2+4x+4)}$ is	**(R)**	$(1,2)\cup(3,\infty)$
(D)	Let $f(x)=\begin{cases} x^2 & x<1 \\ x+1 & x\geq 1\end{cases}$; $g(x)=\begin{cases} x+2 & x<1 \\ x^2 & x\geq 1\end{cases}$	**(S)**	$[0,\infty)$
	Then range of function $f(g(x))$ is	**(T)**	$(-\infty,-3)\cup(-2,-1)\cup(2,\infty)$

7. Let $f(x)\begin{bmatrix} 1+x; & 0\leq x\leq 2 \\ 3-x; & 2<x\leq 3\end{bmatrix}$;

$g(x)=f(f(x))$:

	Column-I		Column-II
(A)	If domain of $g(x)$ is $[a,b]$ then $b-a$ is	**(P)**	1
(B)	If range of $g(x)$ is $[c,d]$ then $c+d$ is	**(Q)**	2
(C)	$f(f(f(2)))+f(f(f(3)))$, is	**(R)**	3
(D)	$m=$ maximum value of $g(x)$ then $2m-2$ is :	**(S)**	4

Answers

1. A $\to$ R ; B $\to$ S ; C $\to$ P ; D $\to$ Q
2. A $\to$ S ; B $\to$ P ; C $\to$ Q ; D $\to$ R
3. A $\to$ Q ; B $\to$ R ; C $\to$ P ; D $\to$ S
4. A $\to$ P, Q, S, T ; B $\to$ Q, R ; C $\to$ P, Q, S ; D $\to$ P, S
5. A $\to$ Q; B $\to$ P; C $\to$ S; D $\to$ R
6. A $\to$ R; B $\to$ P; C $\to$ T; D $\to$ S
7. A $\to$ R ; B $\to$ R ; C $\to$ R ; D $\to$ S

Exercise-5 : Subjective Type Problems

1. Let $f(x)$ be a polynomial of degree 6 with leading coefficient 2009. Suppose further, that $f(1)=1, f(2)=3, f(3)=5, f(4)=7, f(5)=9, f'(2)=2$, then the sum of all the digits of $f(6)$ is
2. Let $f(x)=x^3-3x+1$. Find the number of different real solution of the equation $f(f(x))=0$.
3. If $f(x+y+1)=(\sqrt{f(x)}+\sqrt{f(y)})^2\ \forall\ x, y\in R$ and $f(0)=1$, then $f(2)=$......
4. If the domain of $f(x)=\sqrt{12-3^x-3^{3-x}}+\sin^{-1}\left(\frac{2x}{3}\right)$ is $[a, b]$, then $a=$......
5. The number of elements in the range of the function :
$y=\sin^{-1}\left[x^2+\frac{5}{9}\right]+\cos^{-1}\left[x^2-\frac{4}{9}\right]$ where $[\cdot]$ denotes the greatest integer function is
6. The number of solutions of the equation $f(x-1)+f(x+1)=\sin\alpha,\ 0<\alpha<\frac{\pi}{2}$, where $f(x)=\begin{cases}1-|x| & , |x|\le 1\\ 0 & , |x|>1\end{cases}$ is
7. The number of integers in the range of function $f(x)=[\sin x]+[\cos x]+[\sin x+\cos x]$ is (where $[\cdot]$ = denotes greatest integer function)
8. If $P(x)$ is a polynomial of degree 4 such that $P(-1)=P(1)=5$ and $P(-2)=P(0)=P(2)=2$, then find the maximum value of $P(x)$.
9. The number of integral value(s) of k for which the curve $y=\sqrt{-x^2-2x}$ and $x+y-k=0$ intersect at 2 distinct points is/are
10. Let the solution set of the equation :
$$\sqrt{\left[x+\left[\frac{x}{2}\right]\right]}+\left[\sqrt{\{x\}}+\left[\frac{x}{3}\right]\right]=3$$
is $[a, b)$. Find the product ab.
(where $[\cdot]$ and $\{\cdot\}$ denote greatest integer and fractional part function respectively).
11. For all real number x, let $f(x)=\frac{1}{\sqrt[2011]{1-x^{2011}}}$. Find the number of real roots of the equation
$$f(f(\ldots\ldots(f(x))\ldots\ldots))=\{-x\}$$
where f is applied 2013 times and $\{\cdot\}$ denotes fractional part function.
12. Find the number of elements contained in the range of the function $f(x)=\left[\frac{x}{6}\right]\left[\frac{-6}{x}\right]\ \forall\ x\in(0, 30]$ (where $[\cdot]$ denotes greatest integer function)
13. Let $f(x, y)=x^2-y^2$ and $g(x, y)=2xy$.
such that $(f(x, y))^2-(g(x, y))^2=\frac{1}{2}$ and $f(x, y)\cdot g(x, y)=\frac{\sqrt{3}}{4}$
Find the number of ordered pairs (x, y)?

14. Let $f(x) = \dfrac{x+5}{\sqrt{x^2+1}}$ $\forall\, x \in R$, then the smallest integral value of k for which $f(x) \le k$ $\forall\, x \in R$ is

15. In the above problem, $f(x)$ is injective in the interval $x \in (-\infty, a]$, and λ is the largest possible value of a, then $[\lambda] =$
(where $[x]$ denote greatest integer $\le x$)

16. The number of integral values of m for which $f: R \to R; f(x) = \dfrac{x^3}{3} + (m-1)x^2 + (m+5)x + n$ is bijective is :

17. The number of roots of equation :

$$\left(\frac{(x-1)(x-3)}{(x-2)(x-4)} - e^x\right)\left(\frac{(x+1)(x+3)e^x}{(x+2)(x+4)} - 1\right)(x^3 - \cos x) = 0$$

18. The number of solutions of the equation $\cos^{-1}\left(\dfrac{1-x^2-2x}{(x+1)^2}\right) = \pi(1-\{x\})$, for $x \in [0, 76]$ is equal to. (where $\{\cdot\}$ denote fraction part function)

19. Let $f(x) = x^2 - bx + c$, b is an odd positive integer. Given that $f(x) = 0$ has two prime numbers as roots and $b + c = 35$. If the least value of $f(x)$ $\forall\, x \in R$ is λ, then $\left[\left|\dfrac{\lambda}{3}\right|\right]$ is equal to
(where $[\cdot]$ denotes greatest integer function)

20. Let $f(x)$ be continuous function such that $f(0) = 1$ and $f(x) - f\left(\dfrac{x}{7}\right) = \dfrac{x}{7}$ $\forall\, x \in R$, then $f(42) =$

21. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \min\{f(t) : 0 \le t \le x\} & ; \ 0 \le x \le 1 \\ 3 - x & ; \ 1 < x \le 2 \end{cases}$ and if $\lambda = g\left(\dfrac{1}{4}\right) + g\left(\dfrac{3}{4}\right) + g\left(\dfrac{5}{4}\right)$, then $2\lambda =$

22. If $x = 10\displaystyle\sum_{r=3}^{100} \frac{1}{(r^2-4)}$, then $[x] =$
(where $[\cdot]$ denotes greatest integer function)

23. Let $f(x) = \dfrac{ax+b}{cx+d}$, where a, b, c, d are non zero. If $f(7) = 7$, $f(11) = 11$ and $f(f(x)) = x$ for all x except $-\dfrac{d}{c}$. The unique number which is not in the range of f is

24. Let $A = \{x \mid x^2 - 4x + 3 < 0, x \in R\}$
$B = \{x \mid 2^{1-x} + p \le 0; x^2 - 2(p+7)x + 5 \le 0\}$
If $A \subseteq B$, then the range of real number $p \in [a, b]$ where a, b are integers.
Find the value of $(b - a)$.

25. Let the maximum value of expression $y = \dfrac{x^4 - x^2}{x^6 + 2x^3 - 1}$ for $x > 1$ is $\dfrac{p}{q}$, where p and q are relatively prime natural numbers, then $p + q =$

26. If $f(x)$ is an even function, then the number of distinct real numbers x such that $f(x) = f\left(\dfrac{x+1}{x+2}\right)$ is :

27. The least integral value of $m, m \in R$ for which the range of function $f(x) = \dfrac{x+m}{x^2+1}$ contains the interval $[0, 1]$ is :

28. Let x_1, x_2, x_3 satisfying the equation $x^3 - x^2 + \beta x + \gamma = 0$ are in G.P. where x_1, x_2, x_3 are positive numbers. Then the maximum value of $[\beta] + [\gamma] + 4$ is where $[\cdot]$ denotes greatest integer function is :

29. Let $A = \{1, 2, 3, 4\}$ and $B = \{0, 1, 2, 3, 4, 5\}$. If 'm' is the number of strictly increasing function f, $f : A \to B$ and n is the number of onto functions $g, g : B \to A$. Then the last digit of $n - m$ is.

30. If $\sum_{r=1}^{n} [\log_2 r] = 2010$, where $[\cdot]$ denotes greatest integer function, then the sum of the digits of n is :

31. Let $f(x) = \dfrac{ax+b}{cx+d}$, where a, b, c, d are non-zero. If $f(7) = 7$, $f(11) = 11$ and $f(f(x)) = x$ for all x except $-\dfrac{d}{c}$. The unique number which is not in the range of f is

32. It is pouring down rain, and the amount of rain hitting point (x, y) is given by $f(x, y) = |x^3 + 2x^2y - 5xy^2 - 6y^3|$. If Mr. '$A$' starts at $(0, 0)$; find number of possible value(s) for 'm' such that $y = mx$ is a line along which Mr. 'A' could walk without any rain falling on him.

33. Let $P(x)$ be a cubic polynomical with leading co-efficient unity. Let the remainder when $P(x)$ is divided by $x^2 - 5x + 6$ equals 2 times the remainder when $P(x)$ is divided by $x^2 - 5x + 4$. If $P(0) = 100$, find the sum of the digits of $P(5)$:

34. Let $f(x) = x^2 + 10x + 20$. Find the number of real solution of the equation $f(f(f(f(x)))) = 0$

35. If range of $f(x) = \dfrac{(\ln x)(\ln x^2) + \ln x^3 + 3}{\ln^2 x + \ln x^2 + 2}$ can be expressed as $\left[\dfrac{a}{b}, \dfrac{c}{d}\right]$ where a, b, c and d are prime numbers (not necessarily distinct) then find the value of $\dfrac{(a+b+c+d)}{2}$.

36. Polynomial $P(x)$ contains only terms of odd degree. When $P(x)$ is divided by $(x - 3)$, then remainder is 6. If $P(x)$ is divided by $(x^2 - 9)$ then remainder is $g(x)$. Find the value of $g(2)$.

37. The equation $2x^3 - 3x^2 + p = 0$ has three real roots. Then find the minimum value of p.

38. Find the number of integers in the domain of $f(x) = \dfrac{1}{\sqrt{\ln \cos^{-1} x}}$.

Answers

1.	26	**2.**	7	**3.**	9	**4.**	1	**5.**	1	**6.**	4	**7.**	5
8.	6	**9.**	1	**10.**	12	**11.**	1	**12.**	6	**13.**	4	**14.**	6
15.	0	**16.**	6	**17.**	7	**18.**	76	**19.**	6	**20.**	8	**21.**	5
22.	5	**23.**	9	**24.**	3	**25.**	7	**26.**	4	**27.**	1	**28.**	3
29.	5	**30.**	8	**31.**	9	**32.**	3	**33.**	2	**34.**	2	**35.**	6
36.	4	**37.**	0	**38.**	2								

❑❑❑

Chapter 2 – Limit

2

LIMIT

Exercise-1 : Single Choice Problems

1. $\lim_{x\to 0} \frac{\cos(\tan x)-\cos x}{x^4} =$

(a) $\frac{1}{6}$ (b) $-\frac{1}{3}$ (c) $-\frac{1}{6}$ (d) $\frac{1}{3}$

2. The value of $\lim_{x\to 0} \frac{(\sin x-\tan x)^2-(1-\cos 2x)^4+x^5}{7(\tan^{-1} x)^7+(\sin^{-1} x)^6+3\sin^5 x}$ equal to :

(a) 0 (b) 1 (c) 2 (d) $\frac{1}{3}$

3. Let $a=\lim_{x\to 0}\frac{\ln(\cos 2x)}{3x^2}$, $b=\lim_{x\to 0}\frac{\sin^2 2x}{x(1-e^x)}$, $c=\lim_{x\to 1}\frac{\sqrt{x}-x}{\ln x}$.

Then a, b, c satisfy :

(a) $a<b<c$ (b) $b<c<a$ (c) $a<c<b$ (d) $b<a<c$

4. If $f(x)=\cot^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ and $g(x)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, then $\lim_{x\to a}\frac{f(x)-f(a)}{g(x)-g(a)}, 0<a<\frac{1}{2}$ is :

(a) $\frac{3}{2(1+a^2)}$ (b) $\frac{3}{2}$ (c) $\frac{-3}{2(1+a^2)}$ (d) $-\frac{3}{2}$

5. $\lim_{x\to 0}\left(\frac{(1+x)^{\frac{2}{x}}}{e^2}\right)^{\frac{4}{\sin x}}$ is :

(a) e^4 (b) e^{-4} (c) e^8 (d) e^{-8}

6. $\lim_{x\to\infty}\frac{3}{x}\left[\frac{x}{4}\right]=\frac{p}{q}$ (where [·] denotes greatest integer function), then $p+q$ (where p, q are relative prime) is :

(a) 2 (b) 7 (c) 5 (d) 6

7. $f(x) = \lim_{n\to\infty} \dfrac{x^n + \left(\frac{\pi}{3}\right)^n}{x^{n-1} + \left(\frac{\pi}{3}\right)^{n-1}}$, ($n$ is an even integer), then which of the following is incorrect ?

(a) If $f:\left[\frac{\pi}{3},\infty\right) \to \left[\frac{\pi}{3},\infty\right)$, then function is invertible

(b) $f(x) = f(-x)$ has infinite number of solutions

(c) $f(x) = |f(x)|$ has infinite number of solutions

(d) $f(x)$ is one-one function for all $x \in R$

8. $\lim_{x\to 0} \dfrac{\sin(\pi\cos^2(\tan(\sin x)))}{x^2} =$

(a) π (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) none of these

9. If $f(x) = \begin{cases} \dfrac{(e^{(x+3)\ln 27})^{\frac{x}{27}} - 9}{3^x - 27} & ; \; x < 3 \\ \lambda \dfrac{1-\cos(x-3)}{(x-3)\tan(x-3)} & ; \; x > 3 \end{cases}$

If $\lim_{x\to 3} f(x)$ exist, then $\lambda =$

(a) $\frac{9}{2}$ (b) $\frac{2}{9}$ (c) $\frac{2}{3}$ (d) none of these

10. $\lim_{x\to\frac{\pi}{3}} \dfrac{\sin\left(\frac{\pi}{3} - x\right)}{2\cos x - 1}$ is equal to :

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) $\frac{1}{2}$

11. $\lim_{x\to\frac{\pi}{2}} \dfrac{\sin x}{\cos^{-1}\left[\frac{1}{4}(3\sin x - \sin 3x)\right]}$, (where [·] denotes greatest integer function) is :

(a) $\frac{2}{\pi}$ (b) 1 (c) $\frac{4}{\pi}$ (d) does not exist

12. Let f be a continuous function on R such that $f\left(\frac{1}{4^n}\right) = (\sin e^n)e^{-n^2} + \frac{n^2}{n^2+1}$, then $f(0) =$

(a) 1 (b) 0 (c) -1 (d) $\frac{1}{4}$

13. $\lim\limits_{x\to I^-} \dfrac{e^{\{x\}} - \{x\} - 1}{\{x\}^2}$ equals, where $\{\cdot\}$ is fractional part function and I is an integer, to :

(a) $\dfrac{I}{2}$ (b) $e-2$ (c) I (d) does not exist

14. $\lim\limits_{x\to\infty} (e^{11x} - 7x)^{\frac{1}{3x}}$ is equal to :

(a) $\dfrac{11}{3}$ (b) $\dfrac{3}{11}$ (c) $e^{\frac{3}{11}}$ (d) $e^{\frac{11}{3}}$

15. The value of $\lim\limits_{x\to 0}\left[(1-2x)^n \sum\limits_{r=0}^{n} {}^nC_r \left(\dfrac{x+x^2}{1-2x}\right)^r\right]^{1/x}$ is :

(a) e^n (b) e^{-n} (c) e^{3n} (d) e^{-3n}

16. For a certain value of 'c', $\lim\limits_{x\to\infty} [(x^5 + 7x^4 + 2)^c - x]$ is finite and non-zero. Then the value of limit is :

(a) $\dfrac{7}{5}$ (b) 1 (c) $\dfrac{2}{5}$ (d) None of these

17. The number of non-negative integral values of n for which $\lim\limits_{x\to 0} \dfrac{(\cos x - 1)(\cos x - e^x)}{x^n} = 0$ is :

(a) 1 (b) 2 (c) 3 (d) 4

18. The value of $\lim\limits_{x\to 0}\left(\dfrac{\sin x}{x}\right)^{\frac{1}{1-\cos x}}$:

(a) $e^{-1/3}$ (b) $e^{1/3}$ (c) $e^{-1/6}$ (d) $e^{1/6}$

19. If $\lim\limits_{x\to\infty} (\sqrt{x^2 - x + 1} - ax - b) = 0$, then for $k \ge 2, (k \in N)$ $\lim\limits_{n\to\infty} \sec^{2n}(k!\pi b) =$

(a) a (b) $-a$ (c) $2a$ (d) b

20. If f is a positive function such that $f(x+T) = f(x)\,(T > 0)$, $\forall\, x \in R$, then

$$\lim_{n\to\infty} n\left(\frac{f(x+T) + 2f(x+2T) + \ldots\ldots + nf(x+nT)}{f(x+T) + 4f(x+4T) + \ldots\ldots + n^2 f(x+n^2T)}\right) =$$

(a) 2 (b) $\dfrac{2}{3}$ (c) $\dfrac{3}{2}$ (d) None of these

21. Let $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$

$$265\left(\lim_{h\to 0} \frac{h^4 + 3h^2}{(f(1-h) - f(1))\sin 5h}\right) =$$

(a) 1 (b) 2 (c) 3 (d) -3

22. $\lim_{x\to 0}\left(\frac{\cos x - \sec x}{x^2(x+1)}\right) =$

(a) 0 (b) $-\frac{1}{2}$ (c) -1 (d) -2

23. Let $f(x)$ be a continuous and differentiable function satisfying $f(x+y) = f(x)f(y)\ \forall\ x, y \in R$ if $f(x)$ can be expressed as $f(x) = 1 + xP(x) + x^2Q(x)$ where $\lim_{x\to 0} P(x) = a$ and $\lim_{x\to 0} Q(x) = b$, then $f'(x)$ is equal to :

(a) $a\,f(x)$ (b) $b\,f(x)$

(c) $(a+b)\,f(x)$ (d) $(a+2b)\,f(x)$

24. $\lim_{x\to\frac{\pi}{2}} \frac{\left(1-\tan\frac{x}{2}\right)(1-\sin x)}{\left(1+\tan\frac{x}{2}\right)(\pi-2x)^3} =$

(a) not exist (b) $\frac{1}{8}$ (c) $\frac{1}{16}$ (d) $\frac{1}{32}$

25. $\lim_{x\to\infty}\left(\frac{x-3}{x+2}\right)^x$ is equal to :

(a) e (b) e^{-1} (c) e^{-5} (d) e^5

26. $\lim_{x\to\frac{\pi}{2}} (\cos x)^{\cos x}$ is :

(a) 1 (b) 0 (c) $\frac{1}{e}$ (d) $\frac{2}{e}$

27. If $\lim_{x\to c^-}\{\ln x\}$ and $\lim_{x\to c^+}\{\ln x\}$ exists finitely but they are not equal (where $\{\cdot\}$ denotes fractional part function), then :

(a) 'c' can take only rational values

(b) 'c' can take only irrational values

(c) 'c' can take infinite values in which only one is irrational

(d) 'c' can take infinite values in which only one is rational

28. $\lim_{x\to 0}\left(1+\frac{a\sin bx}{\cos x}\right)^{\frac{1}{x}}$, where a, b are non-zero constants is equal to :

(a) $e^{a/b}$ (b) ab

(c) e^{ab} (d) $e^{b/a}$

29. The value of $\lim\limits_{x\to 0}\left((\cos x)^{\frac{1}{\sin^2 x}} + \dfrac{\sin 2x + 2\tan^{-1} 3x + 3x^2}{\ln(1+3x+\sin^2 x) + xe^x}\right)$ is :

(a) $\sqrt{e}+\dfrac{3}{2}$ (b) $\dfrac{1}{\sqrt{e}}+\dfrac{3}{2}$ (c) $\sqrt{e}+2$ (d) $\dfrac{1}{\sqrt{e}}+2$

30. Let $a=\lim\limits_{x\to 1}\left(\dfrac{x}{\ln x}-\dfrac{1}{x\ln x}\right)$; $b=\lim\limits_{x\to 0}\dfrac{x^3-16x}{4x+x^2}$; $c=\lim\limits_{x\to 0}\dfrac{\ln(1+\sin x)}{x}$ and

$d=\lim\limits_{x\to -1}\dfrac{(x+1)^3}{3[\sin(x+1)-(x+1)]}$, then the matrix $\begin{bmatrix} a & b \\ c & d\end{bmatrix}$ is :

(a) Idempotent (b) Involutary
(c) Non-singular (d) Nilpotent

31. The integral value of n so that $\lim\limits_{x\to 0} f(x)$ where $f(x)=\dfrac{(\sin x - x)\left(2\sin x - \ln\left(\dfrac{1+x}{1-x}\right)\right)}{x^n}$ is a finite non-zero number, is :

(a) 2 (b) 4 (c) 6 (d) 8

32. Consider the function $f(x)=\begin{cases}\dfrac{\max\left(x,\dfrac{1}{x}\right)}{\min\left(x,\dfrac{1}{x}\right)}, & \text{if } x\neq 0\\ 1, & \text{if } x=0\end{cases}$, then $\lim\limits_{x\to 0^-}\{f(x)\}+\lim\limits_{x\to 1^-}\{f(x)\}+\lim\limits_{x\to -1^-}[f(x)]=$

(where $\{\cdot\}$ denotes fraction part function and $[\cdot]$ denotes greatest integer function)

(a) 0 (b) 1 (c) 2 (d) 3

33. $\lim\limits_{x\to\left(\frac{1}{\sqrt{2}}\right)^+}\dfrac{\cos^{-1}(2x\sqrt{1-x^2})}{\left(x-\dfrac{1}{\sqrt{2}}\right)}-\lim\limits_{x\to\left(\frac{1}{\sqrt{2}}\right)^-}\dfrac{\cos^{-1}(2x\sqrt{1-x^2})}{\left(x-\dfrac{1}{\sqrt{2}}\right)}=$

(a) $\sqrt{2}$ (b) $2\sqrt{2}$ (c) $4\sqrt{2}$ (d) 0

34. $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left(\sin\dfrac{\pi}{2k}-\cos\dfrac{\pi}{2k}-\sin\left(\dfrac{\pi}{2(k+2)}\right)+\cos\dfrac{\pi}{2(k+2)}\right)=$

(a) 0 (b) 1
(c) 2 (d) 3

35. $\lim\limits_{x\to 0^+}[1+[x]]^{2/x}$, where $[\cdot]$ is greatest integer function, is equal to :

(a) 0 (b) 1
(c) e^2 (d) Does not exist

36. If m and n are positive integers, then $\lim_{x \to 0} \frac{(\cos x)^{1/m} - (\cos x)^{1/n}}{x^2}$ equals to :

(a) $m - n$
(b) $\frac{1}{n} - \frac{1}{m}$
(c) $\frac{m-n}{2mn}$
(d) None of these

37. The value of ordered pair (a, b) such that $\lim_{x \to 0} \frac{x(1 + a\cos x) - b\sin x}{x^3} = 1$, is :

(a) $\left(-\frac{5}{2}, -\frac{3}{2}\right)$
(b) $\left(\frac{5}{2}, \frac{3}{2}\right)$
(c) $\left(-\frac{5}{2}, \frac{3}{2}\right)$
(d) $\left(\frac{5}{2}, -\frac{3}{2}\right)$

38. What is the value of $a + b$, if $\lim_{x \to 0} \frac{\sin(ax) - \ln(e^x \cos x)}{x\sin(bx)} = \frac{1}{2}$?

(a) 1
(b) 2
(c) 3
(d) $-\frac{1}{2}$

39. Let $\alpha = \lim_{n \to \infty} \frac{(1^3 - 1^2) + (2^3 - 2^2) + \ldots + (n^3 - n^2)}{n^4}$, then α is equal to :

(a) $\frac{1}{3}$
(b) $\frac{1}{4}$
(c) $\frac{1}{2}$
(d) non existent

40. The value of $\lim_{x \to 0} \frac{\cos(\sin x) - \cos x}{x^4}$ is equal to :

(a) $\frac{1}{5}$
(b) $\frac{1}{6}$
(c) $\frac{1}{4}$
(d) $\frac{1}{12}$

41. The value of ordered pair (a, b) such that $\lim_{x \to 0} \frac{x(1 + a\cos x) - b\sin x}{x^3} = 1$, is :

(a) $\left(-\frac{5}{2}, -\frac{3}{2}\right)$
(b) $\left(\frac{5}{2}, \frac{3}{2}\right)$
(c) $\left(-\frac{5}{2}, \frac{3}{2}\right)$
(d) $\left(\frac{5}{2}, -\frac{3}{2}\right)$

42. Consider the sequence :

$$u_n = \sum_{r=1}^{n} \frac{r}{2^r}, \quad n \geq 1$$

Then the limit of u_n as $n \to \infty$ is :

(a) 1
(b) e
(c) $\frac{1}{2}$
(d) 2

43. The value of $\lim_{x \to 0} \left((\cos x)^{\frac{1}{\sin^2 x}} + \frac{\sin 2x + 2\tan^{-1} 3x + 3x^2}{\ln(1 + 3x + \sin^2 x) + xe^x} \right)$ is :

(a) $\sqrt{e} + \frac{3}{2}$
(b) $\frac{1}{\sqrt{e}} + \frac{3}{2}$
(c) $\sqrt{e} + 2$
(d) $\frac{1}{\sqrt{e}} + 2$

44. For $n \in N$, let $f_n(x) = \tan\frac{x}{2}(1+\sec x)(1+\sec 2x)(1+\sec 4x)\ldots\ldots(1+\sec 2^n x)$, the $\lim\limits_{x\to 0}\frac{f_n(x)}{2x}$ is equal to :

(a) 0 (b) 2^n (c) 2^{n-1} (d) 2^{n+1}

45. The value of $\lim\limits_{x\to\frac{\pi}{4}}(1+[x])^{\frac{1}{ln(\tan x)}}$ is :

(where [.] denotes greatest integer function).

(a) 0 (b) 1 (c) e (d) $\frac{1}{e}$

46. If $\lim\limits_{x\to 0}\frac{\{(a-n)nx-\tan x\}\sin nx}{x^2}=0$, $n \neq 0$ then a is equal to :

(a) 0 (b) $1+\frac{1}{n}$ (c) n (d) $n+\frac{1}{n}$

47. The value of $\lim\limits_{n\to\infty}\left(\frac{n!}{n^n}\right)^{\frac{3n^3+4}{4n^4-1}}$, $n \in N$ is equal to :

(a) $\left(\frac{1}{e}\right)^{3/4}$ (b) $e^{3/4}$ (c) e^{-1} (d) 0

48. The value of $\lim\limits_{x\to\infty}\frac{ax^2+bx+c}{dx+e}$ $(a, b, c, d, e \in R-\{0\})$ depends on the sign of :

(a) a only (b) d only

(c) a and d only (d) a, b and d only

49. Let $f(x)=\lim\limits_{n\to\infty}\tan^{-1}\left(4n^2\left(1-\cos\frac{x}{n}\right)\right)$ and $g(x)=\lim\limits_{n\to\infty}\frac{n^2}{2}\ln\cos\left(\frac{2x}{n}\right)$ then $\lim\limits_{x\to 0}\frac{e^{-2g(x)}-e^{f(x)}}{x^6}$ equals.

(a) $\frac{8}{3}$ (b) $\frac{7}{3}$ (c) $\frac{5}{3}$ (d) $\frac{2}{3}$

50. If $f(x)$ be a cubic polynomial and $\lim\limits_{x\to 0}\frac{\sin^2 x}{f(x)}=\frac{1}{3}$ then $f(1)$ can not be equal to :

(a) 0 (b) -5 (c) 3 (d) -2

51. $\lim\limits_{x\to 0}\frac{2e^{\sin x}-e^{-\sin x}-1}{x^2+2x}$ equals to :

(a) $\frac{3}{2}$ (b) $e^{3/2}$ (c) 2 (d) e^2

52. If $x_1, x_2, x_3, \ldots\ldots, x_n$ are the roots of $x^n + ax + b = 0$, then the value of $(x_1 - x_2)(x_1 - x_3)$ $(x_1 - x_4)\ldots\ldots(x_1 - x_n)$ is equal to :

(a) $nx_1 + b$ (b) $nx_1^{n-1} + a$ (c) nx_1^{n-1} (d) nx_1^{n-1}

53. $\lim\limits_{x\to 0} \dfrac{\sqrt[3]{1+\sin^2 x} - \sqrt[4]{1-2\tan x}}{\sin x + \tan^2 x}$ is equal to :

(a) -1 (b) 1 (c) $\dfrac{1}{2}$ (d) $-\dfrac{1}{2}$

54. If $f(x) = \begin{vmatrix} x\cos x & 2x\sin x & x\tan x \\ 1 & x & 1 \\ 1 & 2x & 1 \end{vmatrix}$, find $\lim\limits_{x\to 0} \dfrac{f(x)}{x^2}$.

(a) 0 (b) 1 (c) -1 (d) Does not exist

Answers

1.	(b)	**2.**	(d)	**3.**	(d)	**4.**	(d)	**5.**	(b)	**6.**	(b)	**7.**	(d)	**8.**	(a)	**9.**	(c)	**10.**	(b)
11.	(a)	**12.**	(a)	**13.**	(b)	**14.**	(d)	**15.**	(b)	**16.**	(a)	**17.**	(c)	**18.**	(a)	**19.**	(a)	**20.**	(c)
21.	(c)	**22.**	(c)	**23.**	(a)	**24.**	(d)	**25.**	(c)	**26.**	(a)	**27.**	(d)	**28.**	(c)	**29.**	(d)	**30.**	(d)
31.	(c)	**32.**	(a)	**33.**	(c)	**34.**	(d)	**35.**	(b)	**36.**	(c)	**37.**	(a)	**38.**	(b)	**39.**	(b)	**40.**	(b)
41.	(a)	**42.**	(d)	**43.**	(d)	**44.**	(c)	**45.**	(b)	**46.**	(d)	**47.**	(a)	**48.**	(c)	**49.**	(a)	**50.**	(c)
51.	(a)	**52.**	(b)	**53.**	(c)	**54.**	(c)												

Exercise-2 : One or More than One Answer is/are Correct

1. If $\lim\limits_{x\to 0}(p\tan qx^2 - 3\cos^2 x + 4)^{1/(3x^2)} = e^{5/3}$; $p, q \in R$ then :

(a) $p=\sqrt{2}, q=\dfrac{1}{2\sqrt{2}}$ (b) $p=\dfrac{1}{\sqrt{2}}, q=2\sqrt{2}$ (c) $p=1, q=2$ (d) $p=2, q=4$

2. $\lim\limits_{x\to\infty} 2\left(\sqrt{25x^2+x}-5x\right)$ is equal to :

(a) $\lim\limits_{x\to 0}\dfrac{2x-\log_e(1+x)^2}{5x^2}$ (b) $\lim\limits_{x\to 0}\dfrac{e^{-x}-1+x}{x^2}$

(c) $\lim\limits_{x\to 0}\dfrac{2(1-\cos x^2)}{5x^4}$ (d) $\lim\limits_{x\to 0}\dfrac{\sin\frac{x}{5}}{x}$

3. Let $\lim\limits_{x\to\infty}(2^x+a^x+e^x)^{1/x}=L$

which of the following statement(s) is(are) correct ?

(a) if $L=a\,(a>0)$, then the range of a is $[e,\infty)$

(b) if $L=2e(a>0)$, then the range of a is $\{2e\}$

(c) if $L=e\,(a>0)$, then the range of a is $(0,e]$

(d) if $L=2a\,(a>1)$, then the range of a is $\left(\dfrac{e}{2},\infty\right)$

4. Let $\tan\alpha\cdot x+\sin\alpha\cdot y=\alpha$ and $\alpha\operatorname{cosec}\alpha\cdot x+\cos\alpha\cdot y=1$ be two variable straight lines, α being the parameter. Let P be the point of intersection of the lines. In the limiting position when $\alpha\to 0$, the point P lies on the line :

(a) $x=2$ (b) $x=-1$ (c) $y+1=0$ (d) $y=2$

5. Let $f:R\to[-1,1]$ be defined as $f(x)=\cos(\sin x)$, then which of the following is(are) correct ?

(a) f is periodic with fundamental period 2π (b) Range of $f=[\cos 1, 1]$

(c) $\lim\limits_{x\to\frac{\pi}{2}}\left(f\left(\dfrac{\pi}{2}-x\right)+f\left(\dfrac{\pi}{2}+x\right)\right)=2$ (d) f is neither even nor odd function

6. Let $f(x)=x+\sqrt{x^2+2x}$ and $g(x)=\sqrt{x^2+2x}-x$, then :

(a) $\lim\limits_{x\to\infty}g(x)=1$ (b) $\lim\limits_{x\to\infty}f(x)=1$ (c) $\lim\limits_{x\to-\infty}f(x)=-1$ (d) $\lim\limits_{x\to\infty}g(x)=-1$

7. Which of the following limits does not exist ?

(a) $\lim\limits_{x\to\infty}\operatorname{cosec}^{-1}\left(\dfrac{x}{x+7}\right)$ (b) $\lim\limits_{x\to 1}\sec^{-1}(\sin^{-1}x)$

(c) $\lim\limits_{x\to 0^+}x^{\frac{1}{x}}$ (d) $\lim\limits_{x\to 0}\left(\tan\left(\dfrac{\pi}{8}+x\right)\right)^{\cot x}$

8. If $f(x) = \lim_{n\to\infty} x\left(\frac{3}{2} + [\cos x]\left(\sqrt{n^2+1} - \sqrt{n^2-3n+1}\right)\right)$ where $[y]$ denotes largest integer $\le y$, then identify the correct statement(s).

(a) $\lim_{x\to 0} f(x) = 0$

(b) $\lim_{x\to \frac{\pi}{2}} f(x) = \frac{3\pi}{4}$

(c) $f(x) = \frac{3x}{2}\ \forall\, x \in \left[0, \frac{\pi}{2}\right]$

(d) $f(x) = 0\ \forall\, x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

9. Let $f: R \to R;\ f(x) = \begin{cases} (-1)^n & \text{if } \quad x = \dfrac{1}{2^{2^n}}, \quad n = 1, 2, 3\ldots\ldots \\ 0 & \text{otherwise} \end{cases}$

then identify the correct statement(s).

(a) $\lim_{x\to 0} f(x) = 0$

(b) $\lim_{x\to 0} f(x)$ does not exist

(c) $\lim_{x\to 0} f(x) f(2x) = 0$

(d) $\lim_{x\to 0} f(x) f(2x)$ does not exist

10. If $\lim_{x\to a} f(x) = \lim_{x\to a} [f(x)]$ ([·] denotes the greatest integer function) and $f(x)$ is non-constant continuous function, then :

(a) $\lim_{x\to a} f(x)$ is an integer

(b) $\lim_{x\to a} f(x)$ is non-integer

(c) $f(x)$ has local maximum at $x = a$

(d) $f(x)$ has local minimum at $x = a$

11. Let $f(x) = \dfrac{\cos^{-1}(1-\{x\})\sin^{-1}(1-\{x\})}{\sqrt{2\{x\}}(1-\{x\})}$ where $\{x\}$ denotes the fractional part of x, then :

(a) $\lim_{x\to 0^+} f(x) = \frac{\pi}{4}$

(b) $\lim_{x\to 0^+} f(x) = \sqrt{2} \lim_{x\to 0^-} f(x)$

(c) $\lim_{x\to 0^-} f(x) = \frac{\pi}{4\sqrt{2}}$

(d) $\lim_{x\to 0^-} f(x) = \frac{\pi}{2\sqrt{2}}$

12. If $\lim_{x\to 0} \dfrac{(\sin(\sin x) - \sin x)}{ax^3 + bx^5 + c} = -\dfrac{1}{12}$, then:

(a) $a = 2$ (b) $a = -2$ (c) $c = 0$ (d) $b \in R$

13. If $f(x) = \lim_{n\to\infty} (n(x^{1/n} - 1))$ for $x > 0$, then which of the following is/are true ?

(a) $f\left(\frac{1}{x}\right) = 0$

(b) $f\left(\frac{1}{x}\right) = \frac{1}{f(x)}$

(c) $f\left(\frac{1}{x}\right) = -f(x)$

(d) $f(xy) = f(x) + f(y)$

14. The value of $\lim_{n\to\infty} \cos^2 (\pi (\sqrt[3]{n^3 + n^2 + 2n}))$ (where $n \in N$):

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{9}$

15. If $\alpha, \beta \in \left(-\frac{\pi}{2}, 0\right)$ such that $(\sin\alpha + \sin\beta) + \frac{\sin\alpha}{\sin\beta} = 0$ and $(\sin\alpha + \sin\beta)\ \frac{\sin\alpha}{\sin\beta} = -1$ and $\lambda = \lim_{n\to\infty} \frac{1 + (2\sin\alpha)^{2n}}{(2\sin\beta)^{2n}}$ then :

(a) $a = -\frac{\pi}{6}$ (b) $\lambda = 2$ (c) $\alpha = -\frac{\pi}{3}$ (d) $\lambda = 1$

16. Let $f(x) = \begin{cases} |x-2| + a^2 - 6a + 9 & , \ x < 2 \\ 5 - 2x & , \ x \geq 2 \end{cases}$

If $\lim_{x\to 2} [f(x)]$ exists, the possible values a can take is/are (where $[\cdot]$ represents the greatest integer function)

(a) 2 (b) $\frac{5}{2}$ (c) 3 (d) $\frac{7}{2}$

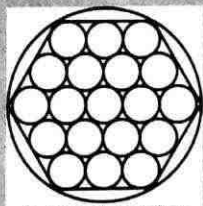
Answers

1.	(b, c)	2.	(a, c, d)	3.	(a, b, c)	4.	(a, c)	5.	(b, c)	6.	(a, c)
7.	(a, d)	8.	(a, c, d)	9.	(b, c)	10.	(a, d)	11.	(b, d)	12.	(a, c)
13.	(c, d)	14.	(c)	15.	(a, b)	16.	(b)				

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

A circular disk of unit radius is filled with a number of smaller circular disks arranged in the form of hexagon. Let A_n denotes a stack of disks arranged in the shape of a hexagon having 'n' disks on a side. The figure shows the configuration A_3. If 'A' be the area of large disk, S_n be the number of disks in A_n configuration and r_n be the radius of each disk in A_n configuration, then



A_3 Configuration

1. $\lim\limits_{n\to\infty} \dfrac{S_n}{n^2}$:

(a) 3 (b) 4 (c) 1 (d) 11

2. $\lim\limits_{n\to\infty} nr_n$:

(a) $\dfrac{1}{3}$ (b) $\dfrac{1}{2}$ (c) $\dfrac{1}{4}$ (d) $\dfrac{1}{11}$

Paragraph for Question Nos. 3 to 4

Let $f(x) = \begin{bmatrix} x+3 & ; & -2 < x < 0 \\ 4 & ; & x = 0 \\ 2x+5 & ; & 0 < x < 1 \end{bmatrix}$, then

3. $\lim\limits_{x\to 0^-} f([x - \tan x])$ is : ([·] denotes greatest integer fucntion)

(a) 2 (b) 4 (c) 5 (d) None of these

4. $\lim\limits_{x\to 0^+} f\left(\left\{\dfrac{x}{\tan x}\right\}\right)$ is : ({·} denotes fractional part of function)

(a) 4 (b) 5 (c) 7 (d) None of these

Paragraph for Question Nos. 5 to 6

A certain function $f(x)$ has the property that $f(3x) = \alpha f(x)$ for all positive real values of x and $f(x) = 1 - |x - 2|$ for $1 \le x \le 3$.

5. $\lim\limits_{x\to 2} (f(x))^{\operatorname{cosec}\left(\frac{\pi x}{2}\right)}$ is :

(a) $\dfrac{2}{\pi}$ (b) $-\dfrac{2}{\pi}$

(c) $e^{2/\pi}$ (d) Non-existent

6. If the total area bounded by $y = f(x)$ and x-axis in $[1, \infty)$ converges to a finite quantity, then the range of α is :

(a) $(-1, 1)$ (b) $\left(-\frac{1}{2}, \frac{1}{2}\right)$ (c) $\left(-\frac{1}{3}, \frac{1}{3}\right)$ (d) $\left(-\frac{1}{4}, \frac{1}{4}\right)$

Paragraph for Question Nos. 7 to 9

Consider the limit $\lim\limits_{x\to 0} \frac{1}{x^3}\left(\frac{1}{\sqrt{1+x}} - \frac{(1+ax)}{(1+bx)}\right)$ exists, finite and has the value equal to l (where a, b are real constants), then :

7. $a =$

(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

8. $a + b =$

(a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

9. $\left|\frac{b}{l}\right| =$

(a) 38 (b) 16 (c) 72 (d) 24

Paragraph for Question Nos. 10 to 11

For the curve $\sin x + \sin y = 1$ lying in the first quadrant there exists a constant α for which

$$\lim_{x\to 0} x^{\alpha} \frac{d^2y}{dx^2} = L \text{ (not zero)}$$

10. The value of α :

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{3}{2}$ (d) 2

11. The value of L :

(a) $\frac{1}{2}$ (b) 1 (c) $\frac{1}{2\sqrt{2}}$ (d) $\frac{1}{2\sqrt{3}}$

Answers

1.	(a)	**2.**	(b)	**3.**	(b)	**4.**	(c)	**5.**	(d)	**6.**	(c)	**7.**	(d)	**8.**	(c)	**9.**	(d)	**10.**	(c)
11.	(c)																		

Exercise-4 : Matching Type Problems

1.

	Column-I		Column-II
(A)	$\lim\limits_{n\to\infty}\left(\dfrac{1+\sqrt[n]{4}}{2}\right)^n =$	(P)	2
(B)	Let $f(x)=\lim\limits_{n\to\infty}\dfrac{2x}{\pi}\tan^{-1}(nx)$, then $\lim\limits_{x\to0^+}f(x)=$	(Q)	0
(C)	$\lim\limits_{x\to\frac{\pi}{2}^+}\dfrac{\cos(\tan^{-1}(\tan x))}{x-\frac{\pi}{2}}=$	(R)	1
(D)	If $\lim\limits_{x\to0^+}(x)^{\frac{1}{\ln\sin x}}=e^L$, then $L+2=$	(S)	3
		(T)	Non-existent

2. [·] represents greatest integer function :

	Column-I		Column-II
(A)	If $f(x)=\sin^{-1}x$ and $\lim\limits_{x\to\frac{1}{2}^+}f(3x-4x^3)=a-3\lim\limits_{x\to\frac{1}{2}^-}f(x)$, then $[a]=$	(P)	2
(B)	If $f(x)=\tan^{-1}g(x)$ where $g(x)=\dfrac{3x-x^3}{1-3x^2}$ and then find $\left[\lim\limits_{h\to0}\dfrac{f\left(\frac{1}{2}+6h\right)-f\left(\frac{1}{2}\right)}{6h}\right]=$	(Q)	3
(C)	If $\cos^{-1}(4x^3-3x)=a+b\cos^{-1}x$ for $-1<x<\dfrac{-1}{2}$, then $[a+b+2]=$	(R)	4
(D)	If $f(x)=\cos^{-1}(4x^3-3x)$ and $\lim\limits_{x\to\frac{1}{2}^+}f'(x)=a$ and $\lim\limits_{x\to\frac{1}{2}^-}f'(x)=b$, then $a+b+3=$	(S)	-2
		(T)	Non existent

Answers

1. A → P ; B → Q ; C → R ; D → S

2. A → Q ; B → P ; C → S ; D → Q

Exercise-5 : Subjective Type Problems

1. If $\lim\limits_{x\to 0} \dfrac{\ln\cot\left(\dfrac{\pi}{4}-\beta x\right)}{\tan\alpha x} = 1$, then $\dfrac{\alpha}{\beta} = \ldots\ldots$

2. If $\lim\limits_{x\to 0} \dfrac{f(x)}{\sin^2 x} = 8$, $\lim\limits_{x\to 0} \dfrac{g(x)}{2\cos x - xe^x + x^3 + x - 2} = \lambda$ and $\lim\limits_{x\to 0} (1+2f(x))^{\frac{1}{g(x)}} = \dfrac{1}{e}$, then $\lambda =$

3. If α, β are two distinct real roots of the equation $ax^3 + x - 1 - a = 0, (a \neq -1, 0)$, none of which is equal to unity, then the value of $\lim\limits_{x\to(1/\alpha)} \dfrac{(1+a)x^3 - x^2 - a}{(e^{1-\alpha x}-1)(x-1)}$ is $\dfrac{al(k\alpha-\beta)}{\alpha}$. Find the value of kl.

4. The value of $\lim\limits_{x\to 0} \dfrac{(140)^x - (35)^x - (28)^x - (20)^x + 7^x + 5^x + 4^x - 1}{x\sin^2 x} = 2\ln 2 \ln k \ln 7$, then $k =$

5. If $\lim\limits_{x\to 0} \dfrac{a\cot x}{x} + \dfrac{b}{x^2} = \dfrac{1}{3}$, then $b - a =$

6. Find the value of $\lim\limits_{x\to\infty} \left(x + \dfrac{1}{x}\right)e^{1/x} - x$.

7. Find $\lim\limits_{x\to\alpha^+} \left[\dfrac{\min(\sin x, \{x\})}{x-1}\right]$ where α is root of equation $\sin x + 1 = x$ (here [$\cdot$] represent greatest integer and {$\cdot$} represent fractional part function)

Answers

1.	2	**2.**	8	**3.**	1	**4.**	5	**5.**	2	**6.**	1	**7.**	0

□□□

Chapter 3 – Continuity, Differentiability and Differentiation

3 CONTINUITY, DIFFERENTIABILITY AND DIFFERENTIATION

Exercise-1 : Single Choice Problems

1. Let 'f' be a differentiable real valued function satisfying $f(x+2y)=f(x)+f(2y)+6xy(x+2y)\ \forall\ x,y\in R$. Then $f''(0), f''(1), f''(2)\ldots\ldots$ are in :

(a) AP (b) GP (c) HP (d) None of these

2. The number of points of non-differentiability for $f(x)=\max\left\{||x|-1|, \frac{1}{2}\right\}$ is :

(a) 4 (b) 3 (c) 2 (d) 5

3. Number of points of discontinuity of $f(x)=\left\{\frac{x}{5}\right\}+\left[\frac{x}{2}\right]$ in $x\in[0,100]$ is/are (where $[\cdot]$ denotes greatest integer function and $\{\cdot\}$ denotes fractional part function)

(a) 50 (b) 51 (c) 52 (d) 61

4. If $f(x)$ has isolated point of discontinuity at $x=a$ such that $|f(x)|$ is continuous at $x=a$ then :

(a) $\lim\limits_{x\to a} f(x)$ does not exist (b) $\lim\limits_{x\to a} f(x)+f(a)=0$

(c) $f(a)=0$ (d) None of these

5. If $f(x)$ is a thrice differentiable function such that, $\lim\limits_{x\to 0}\frac{f(4x)-3f(3x)+3f(2x)-f(x)}{x^3}=12$ then the value of $f'''(0)$ equals to :

(a) 0 (b) 1 (c) 12 (d) None of these

6. $$y=\frac{1}{1+(\tan\theta)^{\sin\theta-\cos\theta}+(\cot\theta)^{\cos\theta-\cot\theta}}+\frac{1}{1+(\tan\theta)^{\cos\theta-\sin\theta}+(\cot\theta)^{\sin\theta-\cot\theta}}$$
$$+\frac{1}{1+(\tan\theta)^{\cos\theta-\cot\theta}+(\cot\theta)^{\cot\theta-\sin\theta}}$$ then $\frac{dy}{dx}$ at $\theta=\pi/3$ is :

(a) 0 (b) 1

(c) $\sqrt{3}$ (d) None of these

7. Let $f'(x)=\sin(x^2)$ and $y=f(x^2+1)$ then $\frac{dy}{dx}$ at $x=1$ is :

(a) $2\sin 2$ (b) $2\cos 2$ (c) $2\sin 4$ (d) $\cos 2$

8. If $f(x) = |\sin x - |\cos x||$, then $f'\left(\frac{7\pi}{6}\right) =$

(a) $\frac{\sqrt{3}+1}{2}$ (b) $\frac{1-\sqrt{3}}{2}$

(c) $\frac{\sqrt{3}-1}{2}$ (d) $\frac{-1-\sqrt{3}}{2}$

9. If $2\sin x \cdot \cos y = 1$, then $\frac{d^2y}{dx^2}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ is ....

(a) -4 (b) -2 (c) -6 (d) 0

10. f is a differentiable function such that $x = f(t^2), y = f(t^3)$ and $f'(1) \neq 0$ if $\left(\frac{d^2y}{dx^2}\right)_{t=1} =$

(a) $\frac{3}{4}\left(\frac{f''(1)+f'(1)}{(f'(1))^2}\right)$ (b) $\frac{3}{4}\left(\frac{f'(1)\cdot f''(1)-f''(1)}{(f'(1))^2}\right)$

(c) $\frac{4}{3}\frac{f''(1)}{(f'(1))^2}$ (d) $\frac{4}{3}\left(\frac{f'(1)f''(1)-f''(1)}{(f'(1))^2}\right)$

11. Let $f(x) = \begin{cases} ax+1 & \text{if } x<1 \\ 3 & \text{if } x=1 \\ bx^2+1 & \text{if } x>1 \end{cases}$. If $f(x)$ is continuous at $x = 1$ then $(a-b)$ is equal to :

(a) 0 (b) 1 (c) 2 (d) 4

12. If $y = 1 + \frac{\alpha}{\left(\frac{1}{x}-\alpha\right)} + \frac{\beta/x}{\left(\frac{1}{x}-\alpha\right)\left(\frac{1}{x}-\beta\right)} + \frac{\gamma/x^2}{\left(\frac{1}{x}-\alpha\right)\left(\frac{1}{x}-\beta\right)\left(\frac{1}{x}-\gamma\right)}$, then $\frac{dy}{dx}$ is :

(a) $y\left(\frac{\alpha}{\alpha-x}+\frac{\beta}{\beta-x}+\frac{\gamma}{\gamma-x}\right)$ (b) $\frac{y}{x}\left(\frac{\alpha}{1/x-\alpha}+\frac{\beta}{1/x-\beta}+\frac{\gamma}{1/x-\gamma}\right)$

(c) $y\left(\frac{\alpha}{1/x-\alpha}+\frac{\beta}{1/x-\beta}+\frac{\gamma}{1/x-\gamma}\right)$ (d) $\frac{y}{x}\left(\frac{\alpha/x}{1/x-\alpha}+\frac{\beta/x}{1/x-\beta}+\frac{\gamma/x}{1/x-\gamma}\right)$

13. If $f(x) = \sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}$; then $f'(0)$ is equal to :

(a) 4 (b) 3 (c) 2 (d) 1

14. Let $f(x) = \begin{cases} \sin^2 x & , x \text{ is rational} \\ -\sin^2 x & , x \text{ is irrational} \end{cases}$, then set of points, where $f(x)$ is continuous, is :

(a) $\left\{(2n+1)\frac{\pi}{2}, n \in I\right\}$ (b) a null set

(c) $\{n\pi, n \in I\}$ (d) set of all rational numbers

15. The number of values of x in $(0, 2\pi)$ where the function $f(x) = \frac{\tan x + \cot x}{2} - \left|\frac{\tan x - \cot x}{2}\right|$ is continuous but non-derivable :

(a) 3 (b) 4 (c) 0 (d) 1

16. If $f(x) = |x-1|$ and $g(x) = f(f(f(x)))$, then $g'(x)$ is equal to :

(a) 1 for $x > 2$ (b) 1 for $2 < x < 3$ (c) -1 for $2 < x < 3$ (d) -1 for $x > 3$

17. If $f(x)$ is a continuous function $\forall\, x \in R$ and the range of $f(x)$ is $(2, \sqrt{26})$ and $g(x) = \left[\frac{f(x)}{C}\right]$ is continuous $\forall\, x \in R$, then the least positive integral value of C is : (where $[\cdot]$ denotes the greatest integer function.)

(a) 3 (b) 5 (c) 6 (d) 7

18. If $y = x + e^x$, then $\left(\frac{d^2x}{dy^2}\right)_{x=\ln 2}$ is :

(a) $-\frac{1}{9}$ (b) $-\frac{2}{27}$ (c) $\frac{2}{27}$ (d) $\frac{1}{9}$

19. Let $f(x) = x^3 + 4x^2 + 6x$ and $g(x)$ be its inverse then the value of $g'(-4)$:

(a) -2 (b) 2 (c) $\frac{1}{2}$ (d) None of these

20. If $f(x) = 2 + |x| - |x-1| - |x+1|$, then $f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) + f'\left(\frac{5}{2}\right)$ is equal to :

(a) 1 (b) -1 (c) 2 (d) -2

21. If $f(x) = \cos(x^2 - 4[x]); 0 < x < 1$, (where $[\cdot]$ denotes greatest integer function) then $f'\left(\frac{\sqrt{\pi}}{2}\right)$ is equal to :

(a) $-\sqrt{\frac{\pi}{2}}$ (b) $\sqrt{\frac{\pi}{2}}$ (c) 0 (d) $\sqrt{\frac{\pi}{4}}$

22. Let $g(x)$ be the inverse of $f(x)$ such that $f'(x) = \frac{1}{1+x^5}$, then $\frac{d^2(g(x))}{dx^2}$ is equal to :

(a) $\frac{1}{1+(g(x))^5}$ (b) $\frac{g'(x)}{1+(g(x))^5}$

(c) $5(g(x))^4(1+(g(x))^5)$ (d) $1+(g(x))^5$

23. Let $f(x) = \begin{cases} \min(x, x^2) & x \geq 0 \\ \max(2x, x-1) & x < 0 \end{cases}$, then which of the following is not true ?

(a) $f(x)$ is not differentiable at $x = 0$

(b) $f(x)$ is not differentiable at exactly two points

(c) $f(x)$ is continuous everywhere

(d) $f(x)$ is strictly increasing $\forall x \in R$

24. If $f(x) = \lim_{n\to\infty}\left(\prod_{i=1}^{n}\cos\left(\frac{x}{2^i}\right)\right)$ then $f'(x)$ is equal to :

(a) $\frac{\sin x}{x}$ (b) $\frac{x}{\sin x}$ (c) $\frac{x\cos x - \sin x}{x^2}$ (d) $\frac{\sin x - x\cos x}{\sin^2 x}$

25. Let $f(x) = \begin{cases} \frac{1-\tan x}{4x-\pi} & x \neq \frac{\pi}{4} \\ \lambda & x = \frac{\pi}{4} \end{cases}; x \in \left[0, \frac{\pi}{2}\right).$

If $f(x)$ is continuous in $\left[0, \frac{\pi}{2}\right)$ then λ is equal to :

(a) 1 (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) -1

26. Let $f(x) = \begin{cases} e^{-\frac{1}{x^2}}\sin\frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$, then $f'(0) =$

(a) 1 (b) -1 (c) 0 (d) Does not exist

27. Let f be a differentiable function satisfying $f'(x) = 2f(x) + 10\ \forall x \in R$ and $f(0) = 0$, then the number of real roots of the equation $f(x) + 5\sec^2 x = 0$ in $(0, 2\pi)$ is :

(a) 0 (b) 1 (c) 2 (d) 3

28. If $f(x) = \begin{cases} \frac{\sin\{\cos x\}}{x - \frac{\pi}{2}} & x \neq \frac{\pi}{2} \\ 1 & x = \frac{\pi}{2} \end{cases}$, where $\{k\}$ represents the fractional part of k, then :

(a) $f(x)$ is continuous at $x = \frac{\pi}{2}$

(b) $\lim_{x\to\frac{\pi}{2}} f(x)$ does not exist

(c) $\lim_{x\to\frac{\pi}{2}} f(x)$ exists, but f is not continuous at $x = \frac{\pi}{2}$

(d) $\lim_{x\to\frac{\pi}{2}^-} f(x) = 1$

29. Let $f(x)$ be a polynomial in x. The second derivative of $f(e^x)$ w.r.t. x is :

(a) $f''(e^x)e^x + f'(e^x)$ (b) $f''(e^x)e^{2x} + f'(e^x)e^{2x}$

(c) $f''(e^x)e^x + f'(e^x)e^{2x}$ (d) $f''(e^x)e^{2x} + e^x f'(e^x)$

30. If $e^{f(x)} = \log_e x$ and $g(x)$ is the inverse function of $f(x)$, then $g'(x)$ is equal to :

(a) $e^x + x$ (b) $e^{e^{e^x}} e^{e^x} e^x$ (c) $e^{e^x + x}$ (d) e^{e^x}

31. If $y = f(x)$ is differentiable $\forall\, x \in R$, then

(a) $y = |f(x)|$ is differentiable $\forall\, x \in R$

(b) $y = f^2(x)$ is non-differentiable for atleast one x

(c) $y = f(x)\,|f(x)|$ is non-differentiable for atleast one x

(d) $y = |f(x)|^3$ is differentiable $\forall\, x \in R$

32. If $f(x) = (x-1)^4 (x-2)^3 (x-3)^2$ then the value of $f'''(1) + f''(2) + f'(3)$ is :

(a) 0 (b) 1 (c) 2 (d) 6

33. If $f(x) = \left(\dfrac{x}{2}\right) - 1$, then on the interval $[0, \pi]$:

(a) $\tan(f(x))$ and $\dfrac{1}{f(x)}$ are both continuous

(b) $\tan(f(x))$ and $\dfrac{1}{f(x)}$ are both discontinuous

(c) $\tan(f(x))$ and $f^{-1}(x)$ are both continuous

(d) $\tan f(x)$ is continuous but $f^{-1}(x)$ is not

34. Let $f(x) = \begin{cases} \dfrac{e^{\frac{1}{x-2}} - 3}{3^{\frac{1}{x-2}} + 1} & x > 2 \\ \dfrac{b \sin\{-x\}}{\{-x\}} & x < 2 \\ c & x = 2 \end{cases}$, where $\{\cdot\}$ denotes fraction part function, is continuous at $x = 2$,

then $b + c =$

(a) 0 (b) 1 (c) 2 (d) 4

35. Let $f(x) = \dfrac{e^{\tan x} - e^x + \ln(\sec x + \tan x) - x}{\tan x - x}$ be a continuous function at $x = 0$. The value of $f(0)$ equals :

(a) $\dfrac{1}{2}$ (b) $\dfrac{2}{3}$ (c) $\dfrac{3}{2}$ (d) 2

36. Let $f(x) = \begin{cases} (1+ax)^{1/x} & x<0 \\ b & x=0 \\ \dfrac{(x+c)^{1/3}-1}{(x+1)^{1/2}-1} & x>0 \end{cases}$, is continuous at $x = 0$, then $3(e^a + b + c)$ is equal to :

(a) 3 (b) 6 (c) 7 (d) 8

37. If $\sqrt{x+y} + \sqrt{y-x} = 5$, then $\dfrac{d^2y}{dx^2} =$

(a) $\dfrac{2}{5}$ (b) $\dfrac{4}{25}$ (c) $\dfrac{2}{25}$ (d) $\dfrac{1}{25}$

38. If $f(x) = x^3 + x^4 + \log x$ and g is the inverse of f, then $g'(2)$ is :

(a) 8 (b) $\dfrac{1}{8}$ (c) 2 (d) $\dfrac{1}{4}$

39. The number of points at which the function,

$$f(x) = \begin{cases} \min\{|x|, x^2\} & \text{if } x \in (-\infty, 1) \text{ otherwise} \\ \min\{2x-1, x^2\} \end{cases}$$

is not differentiable is :

(a) 0 (b) 1 (c) 2 (d) 3

40. If $f(x)$ is a function such that $f(x) + f''(x) = 0$ and $g(x) = (f(x))^2 + (f'(x))^2$ and $g(3) = 8$, then $g(8) =$

(a) 0 (b) 3 (c) 5 (d) 8

41. Let f is twice differentiable on R such that $f(0) = 1, f'(0) = 0$ and $f''(0) = -1$, then for $a \in R$,

$$\lim_{x\to\infty}\left(f\left(\frac{a}{\sqrt{x}}\right)\right)^x =$$

(a) e^{-a^2} (b) $e^{-\frac{a^2}{4}}$ (c) $e^{-\frac{a^2}{2}}$ (d) e^{-2a^2}

42. Let $f_1(x) = e^x$ and $f_{n+1}(x) = e^{f_n(x)}$ for any $n \geq 1, n \in N$. Then for any fixed n, the value of $\dfrac{d}{dx} f_n(x)$ equals :

(a) $f_n(x)$ (b) $f_n(x)f_{n-1}(x)\ldots\ldots f_2(x)f_1(x)$

(c) $f_n(x)f_{n-1}(x)$ (d) $f_n(x)f_{n-1}(x)\ldots\ldots f_2(x)f_1(x)e^x$

43. If $y = \tan^{-1}\left(\dfrac{x^{1/3} - a^{1/3}}{1 + x^{1/3}a^{1/3}}\right)$, $x > 0, a > 0$, then $\dfrac{dy}{dx}$ is :

(a) $\dfrac{1}{x^{2/3}(1+x^{2/3})}$ (b) $\dfrac{3}{x^{2/3}(1+x^{2/3})}$ (c) $\dfrac{1}{3x^{2/3}(1+x^{2/3})}$ (d) $\dfrac{1}{3x^{1/3}(1+x^{2/3})}$

44. The value of $k + f(0)$ so that $f(x) = \begin{cases} \dfrac{\sin(4k-1)x}{3x}, & x < 0 \\ \dfrac{\tan(4k+1)x}{5x}, & 0 < x < \dfrac{\pi}{2} \\ 1, & x = 0 \end{cases}$ can be made continuous at $x = 0$ is :

(a) 1 (b) 2 (c) $\frac{5}{4}$ (d) 0

45. If $y = \tan^{-1}\left(\dfrac{x}{1+\sqrt{1-x^2}}\right)$, $|x| \le 1$, then $\dfrac{dy}{dx}$ at $\left(\dfrac{1}{2}\right)$ is :

(a) $\frac{1}{\sqrt{3}}$ (b) 3 (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{2}{\sqrt{3}}$

46. Let $f(x) = \dfrac{e^x x\cos x - x\log_e(1+x) - x}{x^2}$, $x \ne 0$. If $f(x)$ is continuous at $x = 0$, then $f(0)$ is equal to :

(a) 0 (b) 1 (c) -1 (d) 2

47. A function $f(x) = \max(\sin x, \cos x, 1 - \cos x)$ is non-derivable for n values of $x \in [0, 2\pi]$. Then the value of n is :

(a) 2 (b) 1 (c) 3 (d) 4

48. Let g be the inverse function of a differentiable function f and $G(x) = \dfrac{1}{g(x)}$. If $f(4) = 2$ and $f'(4) = \dfrac{1}{16}$, then the value of $(G'(2))^2$ equals to :

(a) 1 (b) 4 (c) 16 (d) 64

49. If $f(x) = \max.\left(x^4, x^2, \dfrac{1}{81}\right)$ $\forall\, x \in [0, \infty)$, then the sum of the square of reciprocal of all the values of x where $f(x)$ is non-differentiable, is equal to :

(a) 1 (b) 81 (c) 82 (d) $\frac{82}{81}$

50. If $f(x)$ is derivable at $x = 2$ such that $f(2) = 2$ and $f'(2) = 4$, then the value of

$$\lim_{h\to 0} \frac{1}{h^2}(\ln(f(2+h^2)) - \ln(f(2-h^2)))$$ is equal to :

(a) 1 (b) 2 (c) 3 (d) 4

51. Let $f(x) = (x^2 - 3x + 2)\,|(x^3 - 6x^2 + 11x - 6)| + \left|\sin\left(x + \dfrac{\pi}{4}\right)\right|$.

Number of points at which the function $f(x)$ is non-differentiable in $[0, 2\pi]$, is :

(a) 5 (b) 4 (c) 3 (d) 2

52. Let f and g be differentiable functions on R (the set of all real numbers) such that $g(1) = 2 = g'(1)$ and $f'(0) = 4$. If $h(x) = f(2xg(x) + \cos\pi x - 3)$ then $h'(1)$ is equal to :

(a) 28 (b) 24 (c) 32 (d) 18

53. If $f(x) = \dfrac{(x+1)^7\sqrt{1+x^2}}{(x^2-x+1)^6}$, then the value of $f'(0)$ is equal to :

(a) 10 (b) 11 (c) 13 (d) 15

54. Statement-1 : The function $f(x) = \lim\limits_{n\to\infty} \dfrac{\log_e(1+x) - x^{2n}\sin(2x)}{1+x^{2n}}$ is discontinuous at $x = 1$.

Statement-2 : L. H. L. = R. H. L. $\neq f(1)$.

(a) Statement-1 is true, Statement-2 is true and Statement-2 is correct explanation for Statement-1

(b) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation for Statement-1

(c) Statement-1 is true, Statement-2 is false

(d) Statement-1 is false, Statement-2 is true

55. If $f(x) = \begin{cases} x & ; \text{ if } x \text{ is rational} \\ 1-x & ; \text{ if } x \text{ is irrational} \end{cases}$, then number of points for $x \in R$, where $y = f(f(x))$ is discontinuous is :

(a) 0 (b) 1 (c) 2 (d) Infinitely many

56. Number of points where $f(x) = \begin{cases} \max(|x^2-x-2|, x^2-3x) & ; \ x \geq 0 \\ \max(\ln(-x), e^x) & ; \ x < 0 \end{cases}$

is non-differentiable will be :

(a) 1 (b) 2 (c) 3 (d) None of these

57. If the function $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \dfrac{x^2}{2} + \dfrac{x^3}{3}$ and $g(x) = f^{-1}(x)$, then the value of $g'\left(\dfrac{-7}{6}\right)$ equals to :

(a) $\dfrac{1}{5}$ (b) $-\dfrac{1}{5}$ (c) $\dfrac{6}{7}$ (d) $-\dfrac{6}{7}$

58. Find k; if possible ; so that

$$f(x) = \begin{cases} \dfrac{\ln(2-\cos 2x)}{\ln^2(1+\sin 3x)} & ; \ x < 0 \\ k & ; \ x = 0 \\ \dfrac{e^{\sin 2x}-1}{\ln(1+\tan 9x)} & ; \ x > 0 \end{cases}$$

is continuous at $x = 0$.

(a) $\dfrac{2}{3}$ (b) $\dfrac{1}{9}$ (c) $\dfrac{2}{9}$ (d) Not possible

59. Let $x = \frac{1+t}{t^3}; y = \frac{3}{2t^2} + \frac{2}{t}$ then the value of $\frac{dy}{dx} - x\left(\frac{dy}{dx}\right)^3$ is :

(a) 2 (b) 0 (c) −1 (d) −2

60. If $y^{-2} = 1 + 2\sqrt{2}\cos 2x$, then :

$$\frac{d^2y}{dx^2} = y\,(py^2+1)(qy^2-1) \text{ then the value of } (p+q) \text{ equals to :}$$

(a) 7 (b) 8 (c) 9 (d) 10

61. Let $f: R \to R$ is not identically zero, differentiable function and satisfy the equations $f(xy) = f(x)f(y)$ and $f(x+z) = f(x) + f(z)$, then $f(5) =$

(a) 3 (b) 5 (c) 10 (d) 15

62. Number of points at which the function $f(x) = \begin{cases} \min.(x, x^2) & \text{if } -\infty < x < 1 \\ \min.(2x-1, x^2) & \text{if } x \geq 1 \end{cases}$ is not derivable is :

(a) 0 (b) 1 (c) 2 (d) 3

63. If $y = (x + \sqrt{1+x^2})^n$, then $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is :

(a) n^2y (b) $-n^2y$ (c) $-y$ (d) $2x^2y$

64. If $g(x) = f\left(x - \sqrt{1-x^2}\right)$ and $f'(x) = 1 - x^2$ then $g'(x)$ equals to :

(a) $1-x^2$ (b) $\sqrt{1-x^2}$ (c) $2x\left(x+\sqrt{1-x^2}\right)$ (d) $2x\left(x-\sqrt{1-x^2}\right)$

65. Let $f(x) = \lim\limits_{n\to\infty} \frac{\log_e(2+x) - x^{2n}\sin x}{1+x^{2n}}$ then :

(a) $f(x)$ is continuous at $x = 1$

(b) $\lim\limits_{x\to 1^-} f(x) = \log_e 3$

(c) $\lim\limits_{x\to 1^+} f(x) = -\sin 1$

(d) $\lim\limits_{x\to 1^+} f(x)$ does not exist

66. Let $f(x+y) = f(x)f(y)$ for all x and y, and $f(5) = -2$, $f'(0) = 3$, then $f'(5)$ is equal to :

(a) 3 (b) 1 (c) −6 (d) 6

67. Let $f(x) = \lim\limits_{n\to\infty} \frac{\log_e(2+x) - x^{2n}\sin x}{1+x^{2n}}$ then :

(a) $f(x)$ is continuous at $x = 1$

(b) $\lim\limits_{x\to 1^+} f(x) = \log_e 3$

(c) $\lim\limits_{x\to 1^+} f(x) = -\sin 1$

(d) $\lim\limits_{x\to 1^-} f(x)$ does not exist

68. If $f(x) = \begin{cases} \dfrac{x - e^x + 1 - \{1 - \cos 2x\}}{x^2} & x \neq 0 \\ k & x = 0 \end{cases}$ is continuous at $x = 0$ then, which of the following statement is false ?

(a) $k = \dfrac{-5}{2}$ (b) $\{k\} = \dfrac{1}{2}$ (c) $[k] = -2$ (d) $[k]\{k\} = \dfrac{-3}{2}$

(where [·] denotes greatest integer function and {·} denotes fraction part function.)

69. Let $f(x) = ||x^2 - 10x + 21| - p|$; then the exhaustive set of values of p for which $f(x)$ has exactly 6 points of non-derivability; is :

(a) $(4, \infty)$ (b) $(0, 4)$ (c) $[0, 4]$ (d) $(-4, 4)$

70. If $f(x) = \sqrt{\dfrac{1 + \sin^{-1} x}{1 - \tan^{-1} x}}$; then $f'(0)$ is equal to :

(a) 4 (b) 3 (c) 2 (d) 1

71. For $t \in (0, 1)$; let $x = \sqrt{2^{\sin^{-1} t}}$ and $y = \sqrt{2^{\cos^{-1} t}}$,

then $1 + \left(\dfrac{dy}{dx}\right)^2$ equals :

(a) $\dfrac{x^2}{y^2}$ (b) $\dfrac{y^2}{x^2}$ (c) $\dfrac{x^2 + y^2}{y^2}$ (d) $\dfrac{x^2 + y^2}{x^2}$

72. Let $f(x) = -1 + |x - 2|$ and $g(x) = 1 - |x|$ then set of all possible value(s) of x for which (fog) (x) is discontinuous is :

(a) $\{0, 1, 2\}$ (b) $\{0, 2\}$ (c) $\{0\}$ (d) an empty set

73. If $f(x) = [x] \tan(\pi x)$ then $f'(K^+)$ is equal to ($k \in I$ and [.] denotes greatest integer function):

(a) $(k-1)\pi(-1)^k$ (b) $k\pi$ (c) $k\pi(-1)^{k+1}$ (d) $(k-1)\pi(-1)^{k+1}$

74. If $f(x) = \begin{cases} \dfrac{ae^{\sin x} + be^{-\sin x} - c}{x^2}; & x \neq 0 \\ 2 \quad ; & x = 0 \end{cases}$ is continuous at $x = 0$; then :

(a) $a = b = c$ (b) $a = 2b = 3c$ (c) $a = b = 2c$ (d) $2a = 2b = c$

75. If $\tan x \cdot \cot y = \sec \alpha$ where α is constant and $\alpha \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$ then $\dfrac{d^2y}{dx^2}$ at $\left(\dfrac{\pi}{4}, \dfrac{\pi}{4}\right)$ equals to :

(a) 0 (b) 1 (c) 2 (d) 3

76. If $y = (x-3)(x-2)(x-1) \times (x+1)(x+2)(x+3)$, then $\dfrac{d^2y}{dx^2}$ at $x = 1$ is :

(a) -101 (b) 48 (c) 56 (d) 190

77. Let $f(x+y)=f(x)f(y)\ \forall x,y \in R, f(0)\neq 0$. If $f(x)$ is continuous at $x=0$, then $f(x)$ is continuous at :

(a) all natural numbers only (b) all integers only

(c) all rational numbers only (d) all real numbers

78. If $f(x)=3x^9-2x^4+2x^3-3x^2+x+\cos x+5$ and $g(x)=f^{-1}(x)$; then the value of $g'(6)$ equals :

(a) 1 (b) $\frac{1}{2}$ (c) 2 (d) 3

79. If $y=f(x)$ and $z=g(x)$ then $\frac{d^2y}{dz^2}$ equals

(a) $\frac{g'f''-f'g''}{(g')^2}$ (b) $\frac{g'f''-f'g''}{(g')^3}$ (c) $\frac{f'g''-g'f''}{(g')^3}$ (d) None of these

80. Let $f(x)=\begin{cases} x+1 & ; x<0 \\ |x-1| & ; x\geq 0\end{cases}$ and $g(x)=\begin{cases} x+1 & ; x<0 \\ (x-1)^2 & ; x\geq 0\end{cases}$ then

the number of points where $g(f(x))$ is not differentiable.

(a) 0 (b) 1 (c) 2 (d) None of these

81. Let $f(x)=[\sin x]+[\cos x]$, $x\in[0,2\pi]$, where $[\cdot]$ denotes the greatest integer function, total number of points where $f(x)$ is non differentiable is equal to :

(a) 2 (b) 3 (c) 4 (d) 5

82. Let $f(x)=\cos x, g(x)=\begin{cases}\min\{f(t):0\leq t\leq x\} & , x\in[0,\pi] \\ (\sin x)-1 & , x>\pi\end{cases}$

Then

(a) $g(x)$ is discontinuous at $x=\pi$ (b) $g(x)$ is continuous for $x\in[0,\infty)$

(c) $g(x)$ is differentiable at $x=\pi$ (d) $g(x)$ is differentiable for $x\in[0,\infty)$

83. If $f(x)=(4+x)^n, n\in N$ and $f^r(0)$ represents the r^{th} derivative of $f(x)$ at $x=0$, then the value of $\sum_{r=0}^{\infty}\frac{f^r(0)}{r!}$ is equal to :

(a) 2^n (b) 3^n (c) 5^n (d) 4^n

84. Let $f(x)=\begin{cases}\frac{x}{1+|x|} & , |x|\geq 1 \\ \frac{x}{1-|x|} & , |x|<1\end{cases}$, then domain of $f'(x)$ is :

(a) $(-\infty,\infty)$ (b) $(-\infty,\infty)-\{-1,0,1\}$ (c) $(-\infty,\infty)-\{-1,1\}$ (d) $(-\infty,\infty)-\{0\}$

85. If the function $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ and $g(x) = f^{-1}(x)$, then the value of $g'\left(\frac{-7}{6}\right)$ equals :

(a) $\frac{1}{5}$ (b) $-\frac{1}{5}$ (c) $\frac{6}{7}$ (d) $-\frac{6}{7}$

86. The number of points at which the function $f(x) = (x-|x|)^2(1-x+|x|)^2$ is not differentiable in the interval $(-3, 4)$ is :

(a) Zero (b) One (c) Two (d) Three

87. If $f(x) = \sqrt{\frac{1+\sin^{-1} x}{1-\tan^{-1} x}}$; then $f'(0)$ is equal to :

(a) 4 (b) 3 (c) 2 (d) 1

88. If $f(x) = \begin{cases} e^{x-1} & ; \ 0 \le x \le 1 \\ x+1-\{x\} & ; \ 1 < x < 3 \end{cases}$ and $g(x) = x^2 - ax + b$ such that $f(x)g(x)$ is continuous in $[0, 3)$ then the ordered pair (a, b) is (where $\{\cdot\}$ denotes fractional part function) :

(a) $(2, 3)$ (b) $(1, 2)$ (c) $(3, 2)$ (d) $(2, 2)$

89. Use the following table and the fact that $f(x)$ is invertible and differentiable everywhere to find $f^{-1}(3)$:

x	$f(x)$	$f'(x)$
3	1	7
6	2	10
9	3	5

(a) 0 (b) $\frac{1}{5}$ (c) $\frac{1}{10}$ (d) $\frac{1}{7}$

90. Let $f(x) = \begin{cases} x^n \sin\frac{1}{x} & , \ x \ne 0 \\ 0 & , \ x = 0 \end{cases}$

Such that $f(x)$ is continuous at $x = 0$; $f'(0)$ is real and finite; and $\lim_{x\to 0} f'(x)$ does not exist. This holds true for which of the following values of n ?

(a) 0 (b) 1 (c) 2 (d) 3

Answers

1.	(a)	**2.**	(d)	**3.**	(a)	**4.**	(b)	**5.**	(c)	**6.**	(a)	**7.**	(c)	**8.**	(c)	**9.**	(a)	**10.**	(a)
11.	(a)	**12.**	(b)	**13.**	(d)	**14.**	(c)	**15.**	(b)	**16.**	(c)	**17.**	(c)	**18.**	(b)	**19.**	(c)	**20.**	(d)
21.	(a)	**22.**	(c)	**23.**	(b)	**24.**	(c)	**25.**	(c)	**26.**	(c)	**27.**	(a)	**28.**	(b)	**29.**	(d)	**30.**	(c)
31.	(d)	**32.**	(a)	**33.**	(c)	**34.**	(a)	**35.**	(c)	**36.**	(c)	**37.**	(c)	**38.**	(b)	**39.**	(b)	**40.**	(d)
41.	(c)	**42.**	(b)	**43.**	(c)	**44.**	(b)	**45.**	(a)	**46.**	(a)	**47.**	(c)	**48.**	(a)	**49.**	(c)	**50.**	(d)
51.	(c)	**52.**	(c)	**53.**	(c)	**54.**	(c)	**55.**	(a)	**56.**	(c)	**57.**	(a)	**58.**	(c)	**59.**	(c)	**60.**	(d)
61.	(b)	**62.**	(b)	**63.**	(a)	**64.**	(c)	**65.**	(c)	**66.**	(c)	**67.**	(c)	**68.**	(c)	**69.**	(b)	**70.**	(d)
71.	(d)	**72.**	(d)	**73.**	(b)	**74.**	(d)	**75.**	(a)	**76.**	(c)	**77.**	(d)	**78.**	(a)	**79.**	(b)	**80.**	(c)
81.	(d)	**82.**	(b)	**83.**	(c)	**84.**	(c)	**85.**	(a)	**86.**	(a)	**87.**	(d)	**88.**	(c)	**89.**	(b)	**90.**	(c)

Exercise-2 : One or More than One Answer is/are Correct

1. If $f(x) = \tan^{-1}(\text{sgn}(x^2 - \lambda x + 1))$ has exactly one point of discontinuity, then the value of λ can be :

(a) 1 (b) -1 (c) 2 (d) -2

2. $f(x) = \begin{cases} 2(x+1) & ; \quad x \le -1 \\ \sqrt{1-x^2} & ; \quad -1 < x < 1 \\ \big|\,||x|-1|-1\,\big| & ; \quad x \ge 1 \end{cases}$, then :

(a) $f(x)$ is non-differentiable at exactly three points

(b) $f(x)$ is continuous in $(-\infty, 1]$

(c) $f(x)$ is differentiable in $(-\infty, -1)$

(d) $f(x)$ is finite type of discontinuity at $x = 1$, but continuous at $x = -1$

3. Let $f(x) = \begin{cases} \dfrac{x(3e^{1/x} + 4)}{2 - e^{1/x}} & ; \quad x \ne 0 \quad x \ne \dfrac{1}{\ln 2} \\ 0 & ; \quad x = 0 \end{cases}$

which of the following statement(s) is/are correct ?

(a) $f(x)$ is continuous at $x = 0$

(b) $f(x)$ is non-derivable at $x = 0$

(c) $f'(0^+) = -3$

(d) $f'(0^-)$ does not exist

4. Let $|f(x)| \le \sin^2 x, \forall x \in R$, then

(a) $f(x)$ is continuous at $x = 0$

(b) $f(x)$ is differentiable at $x = 0$

(c) $f(x)$ is continuous but not differentiable at $x = 0$

(d) $f(0) = 0$

5. Let $f(x) = \begin{cases} \dfrac{a(1 - x\sin x) + b\cos x + 5}{x^2} & ; \quad x < 0 \\ 3 & ; \quad x = 0 \\ \left(1 + \left(\dfrac{cx + dx^3}{x^2}\right)\right)^{\frac{1}{x}} & ; \quad x > 0 \end{cases}$

If f is continuous at $x = 0$ then correct statement(s) is/are :

(a) $a + c = -1$

(b) $b + c = -4$

(c) $a + b = -5$

(d) $c + d =$ an irrational number

6. If $f(x) = \big|\,||x| - 2| + p\,\big|$ have more than 3 points of non-derivability then the value of p can be :

(a) 0

(b) -1

(c) -2

(d) 2

7. Identify the options having correct statement :

(a) $f(x) = \sqrt[3]{x^2|x|} - 1 - |x|$ is no where non-differentiable

(b) $\lim\limits_{x \to \infty} ((x+5)\tan^{-1}(x+1)) - ((x+1)\tan^{-1}(x+1)) = 2\pi$

(c) $f(x) = \sin(\ln(x + \sqrt{x^2+1}))$ is an odd function

(d) $f(x) = \dfrac{4-x^2}{4x-x^3}$ is discontinuous at exactly one point

8. A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions :

$f(0) = 2$; $f'(0) = -5$ and $f''(0) = 3$.

The function $g(x)$ is defined by $g(x) = e^{ax} + f(x)\ \forall x \in R$, where '$a$' is any constant. If $g'(0) + g''(0) = 0$ then 'a' can be equal to :

(a) 1 (b) −1 (c) 2 (d) −2

9. If $f(x) = |x|\sin x$, then f is :

(a) differentiable everywhere (b) not differentiable at $x = n\pi, n \in I$

(c) not differentiable at $x = 0$ (d) continuous at $x = 0$

10. Let [ ] denotes the greatest integer function and $f(x) = [\tan^2 x]$, then

(a) $\lim\limits_{x \to 0} f(x)$ does not exist (b) $f(x)$ is continuous at $x = 0$

(c) $f(x)$ is not differentiable at $x = 0$ (d) $f'(0) = 0$

11. Let f be a differentiable function satisfying $f'(x) = f'(-x)\ \forall x \in R$. Then

(a) If $f(1) = f(2)$, then $f(-1) = f(-2)$

(b) $\dfrac{1}{2}f(x) + \dfrac{1}{2}f(y) = f\left(\dfrac{1}{2}(x+y)\right)$ for all real values of x, y

(c) Let $f(x)$ be an even function, then $f(x) = 0\ \forall x \in R$

(d) $f(x) + f(-x) = 2f(0)\ \forall x \in R$

12. Let $f: R \to R$ be a function, such that $|f(x)| \le x^{4n}$, $n \in N\ \forall x \in R$ then $f(x)$ is :

(a) discontinuous at $x = 0$ (b) continuous at $x = 0$

(c) non-differentiable at $x = 0$ (d) differentiable at $x = 0$

13. Let $f(x) = [x]$ and $g(x) = 0$ when x is an integer and $g(x) = x^2$ when x is not an integer ([ ] is the greatest integer function) then :

(a) $\lim\limits_{x \to 1} g(x)$ exists, but $g(x)$ is not continuous at $x = 1$

(b) $\lim\limits_{x \to 1} f(x)$ does not exist

(c) gof is continuous for all x

(d) fog is continuous for all x

14. Let the function f be defined by $f(x) = \begin{cases} p + qx + x^2 & , \quad x < 2 \\ 2px + 3qx^2 & , \quad x \geq 2 \end{cases}$. Then :

(a) $f(x)$ is continuous in R if $3p + 10q = 4$

(b) $f(x)$ is differentiable in R if $p = q = \dfrac{4}{13}$

(c) If $p = -2, q = 1$, then $f(x)$ is continuous in R

(d) $f(x)$ is differentiable in R if $2p + 11q = 4$

15. Let $f(x) = |2x - 9| + |2x| + |2x + 9|$. Which of the following are true ?

(a) $f(x)$ is not differentiable at $x = \dfrac{9}{2}$

(b) $f(x)$ is not differentiable at $x = \dfrac{-9}{2}$

(c) $f(x)$ is not differentiable at $x = 0$

(d) $f(x)$ is differentiable at $x = \dfrac{-9}{2}, 0, \dfrac{9}{2}$

16. Let $f(x) = \max(x, x^2, x^3)$ in $-2 \leq x \leq 2$. Then :

(a) $f(x)$ is continuous in $-2 \leq x \leq 2$

(b) $f(x)$ is not differentiable at $x = 1$

(c) $f(-1) + f\left(\dfrac{3}{2}\right) = \dfrac{35}{8}$

(d) $f'(-1) f'\left(\dfrac{3}{2}\right) = \dfrac{-35}{4}$

17. If $f(x)$ be a differentiable function satisfying $f(y) f\left(\dfrac{x}{y}\right) = f(x) \ \forall \ x, y \in R, y \neq 0$ and $f(1) \neq 0$, $f'(1) = 3$, then :

(a) $\operatorname{sgn}(f(x))$ is non-differentiable at exactly one point

(b) $\lim\limits_{x \to 0} \dfrac{x^2(\cos x - 1)}{f(x)} = 0$

(c) $f(x) = x$ has 3 solutions

(d) $f(f(x)) - f^3(x) = 0$ has infinitely many solutions

18. Let $f(x) = (x^2 - 3x + 2)(x^2 + 3x + 2)$ and α, β, γ satisfy $\alpha < \beta < \gamma$ are the roots of $f'(x) = 0$ then which of the following is/are correct ([$\cdot$] denotes greatest integer function) ?

(a) $[\alpha] = -2$

(b) $[\beta] = -1$

(c) $[\beta] = 0$

(d) $[\alpha] = 1$

19. Let the function f be defined by $f(x) = \begin{cases} p + qx + x^2, & x < 2 \\ 2px + 3qx^2, & x \geq 2 \end{cases}$. Then :

(a) $f(x)$ is continuous in R if $3p + 10q = 4$

(b) $f(x)$ is differentiable in R if $p = q = \dfrac{4}{13}$

(c) If $p = -2, q = 1$, then $f(x)$ is continuous in R

(d) $f(x)$ is differentiable in R if $2p + 11q = 4$

20. If $y = e^{x\sin(x^3)} + (\tan x)^x$ then $\dfrac{dy}{dx}$ may be equal to :

(a) $e^{x\sin(x^3)}[3x^3\cos(x^3)+\sin(x^3)]+(\tan x)^x[\ln\tan x+2x\operatorname{cosec}2x]$

(b) $e^{x\sin(x^3)}[x^3\cos(x^3)+\sin(x^3)]+(\tan x)^x[\ln\tan x+2x\operatorname{cosec}2x]$

(c) $e^{x\sin(x^3)}[x^3\sin(x^3)+\cos(x^3)]+(\tan x)^x[\ln\tan x+2\operatorname{cosec}2x]$

(d) $e^{x\sin(x^3)}[3x^3\cos(x^3)+\sin(x^3)]+(\tan x)^x\left[\ln\tan x+\dfrac{x\sec^2 x}{\tan x}\right]$

21. Let $f(x) = x+(1-x)x^2+(1-x)(1-x^2)x^3+\ldots\ldots+(1-x)(1-x^2)\ldots\ldots(1-x^{n-1})x^n;(n\ge 4)$ then :

(a) $f(x) = -\prod_{r=1}^{n}(1-x^r)$

(b) $f(x) = 1-\prod_{r=1}^{n}(1-x^r)$

(c) $f'(x) = (1-f(x))\left(\sum_{r=1}^{n}\dfrac{rx^{r-1}}{(1-x^r)}\right)$

(d) $f'(x) = f(x)\left(\sum_{r=1}^{n}\dfrac{rx^{r-1}}{(1-x^r)}\right)$

22. Let $f(x) = \begin{cases} x^2+a\ ;\ 0\le x<1 \\ 2x+b\ ;\ 1\le x\le 2\end{cases}$ and $g(x) = \begin{cases} 3x+b\ ;\ 0\le x<1 \\ x^3\ ;\ 1\le x\le 2\end{cases}$

If derivative of $f(x)$ w.r.t. $g(x)$ at $x=1$ exists and is equal to λ, then which of the following is/are correct ?

(a) $a+b=-3$ (b) $a-b=1$ (c) $\dfrac{ab}{\lambda}=3$ (d) $\dfrac{-b}{\lambda}=3$

23. If $f(x) = \begin{cases} \dfrac{\sin[x^2]\pi}{x^2-3x+8}+ax^3+b\ ;\ 0\le x\le 1 \\ 2\cos\pi x+\tan^{-1}x\ ;\ 1<x\le 2\end{cases}$ is differentiable in [0, 2] then :

([.] denotes greatest integer function)

(a) $a=\dfrac{1}{3}$ (b) $a=\dfrac{1}{6}$ (c) $b=\dfrac{\pi}{4}-\dfrac{13}{6}$ (d) $b=\dfrac{\pi}{4}-\dfrac{7}{3}$

24. If $f(x) = \begin{cases} 1+x & 0\le x\le 2 \\ 3-x & 2<x\le 3\end{cases}$, then $f(f(x))$ is not differentiable at :

(a) $x=1$ (b) $x=2$ (c) $x=\dfrac{5}{2}$ (d) $x=3$

25. Let $f(x)=(x+1)(x+2)(x+3)\ldots\ldots(x+100)$ and $g(x)=f(x)f''(x)-(f'(x))^2$. Let n be the number of real roots of $g(x)=0$, then :

(a) $n<2$ (b) $n>2$ (c) $n<100$ (d) $n>100$

26. If $f(x)=\begin{cases} |x|-3 & , \quad x<1 \\ |x-2|+a & , \quad x\geq 1 \end{cases}$, $g(x)=\begin{cases} 2-|x| & , \quad x<2 \\ \operatorname{sgn}(x)-b & , \quad x\geq 2 \end{cases}$

If $h(x)=f(x)+g(x)$ is discontinuous at exactly one point, then which of the following are correct ?

(a) $a=-3, b=0$ (b) $a=-3, b=-1$ (c) $a=2, b=1$ (d) $a=0, b=1$

27. Let $f(x)$ be a continuous function in [–1, 1] such that

$$f(x)=\begin{cases} \dfrac{\ln(ax^2+bx+c)}{x^2} & ; -1\leq x<0 \\ 1 & ; \quad x=0 \\ \dfrac{\sin(e^{x^2}-1)}{x^2} & ; \quad 0<x\leq 1 \end{cases}$$

Then which of the following is/are correct ?

(a) $a+b+c=0$ (b) $b=a+c$ (c) $c=1+b$ (d) $b^2+c^2=1$

28. $f(x)$ is differentiable function satisfying the relationship $f^2(x)+f^2(y)+2(xy-1)=f^2(x+y)$ $\forall x, y \in R$

Also $f(x)>0\ \forall x\in R$ and $f(\sqrt{2})=2$. Then which of the following statement(s) is/are correct about $f(x)$?

(a) $[f(3)]=3$ ([·] denotes greatest integer function)

(b) $f(\sqrt{7})=3$

(c) $f(x)$ is even

(d) $f'(0)=0$

29. The function $f(x)=\left[\sqrt{1-\sqrt{1-x^2}}\right]$, (where [·] denotes greatest integer function) :

(a) has domain [–1, 1]

(b) is discontinuous at two points in its domain

(c) is discontinuous at $x=0$

(d) is discontinuous at $x=1$

30. A function $f(x)$ satisfies the relation :

$f(x+y)=f(x)+f(y)+xy(x+y)\ \forall x, y \in R$. If $f'(0)=-1$, then :

(a) $f(x)$ is a polynomial function

(b) $f(x)$ is an exponential function

(c) $f(x)$ is twice differentiable for all $x\in R$

(d) $f'(3)=8$

31. The points of discontinuities of $f(x) = \left[\frac{6x}{\pi}\right]\cos\left[\frac{3x}{\pi}\right]$ in $\left[\frac{\pi}{6}, \pi\right]$ is/are :

(where [·] denotes greatest integer function)

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) π

32. Let $f(x) = \begin{cases} \frac{x^2}{2} & 0 \le x < 1 \\ 2x^2 - 3x + \frac{3}{2} & 1 \le x \le 2 \end{cases}$, then in [0, 2] :

(a) $f(x), f'(x)$ are continuous

(b) $f'(x)$ is continuous, $f''(x)$ is not continuous

(c) $f''(x)$ is continuous

(d) $f''(x)$ is non differentiable

33. If $x = \phi(t), y = \psi(t)$, then $\frac{d^2y}{dx^2} =$

(a) $\frac{\phi'\psi'' - \psi'\phi''}{(\phi')^2}$ (b) $\frac{\phi'\psi'' - \psi'\phi''}{(\phi')^3}$ (c) $\frac{\psi''}{\phi'} - \frac{\psi'\phi''}{(\phi')^2}$ (d) $\frac{\psi''}{(\phi')^2} - \frac{\psi'\phi''}{(\phi')^3}$

34. $f(x) = [x]$ and $g(x) = \begin{cases} 0 & , & x \in I \\ x^2 & , & x \notin I \end{cases}$ where [·] denotes the greatest integer function. Then

(a) *gof* is continuous for all x

(b) *gof* is not continuous for all x

(c) *fog* is continuous everywhere

(d) *fog* is not continuous everywhere

35. Let $f : R^+ \to R$ defined as $f(x) = e^x + \ln x$ and $g = f^{-1}$ then correct statement(s) is/are :

(a) $g''(e) = \frac{1-e}{(1+e)^3}$ (b) $g''(e) = \frac{e-1}{(1+e)^3}$ (c) $g'(e) = e + 1$ (d) $g'(e) = \frac{1}{e+1}$

36. Let $f(x) = \begin{cases} \frac{3x - x^2}{2} & ; & x < 2 \\ [x-1] & ; & 2 \le x < 3 \\ x^2 - 8x + 17 & ; & x \ge 3 \end{cases}$; then which of the following hold(s) good ?

([·] denotes greatest integer function)

(a) $\lim_{x \to 2} f(x) = 1$ (b) $f(x)$ is differentiable at $x = 2$

(c) $f(x)$ is continuous at $x = 2$ (d) $f(x)$ is discontinuus at $x = 3$

Answers

1.	(c, d)	**2.**	(a, c, d)	**3.**	(a, b, c)	**4.**	(a, b, d)	**5.**	(a, b, c, d)	**6.**	(b, c)
7.	(a, b, c)	**8.**	(a, d)	**9.**	(a, d)	**10.**	(b, d)	**11.**	(a, d)	**12.**	(b, d)
13.	(a, b, c)	**14.**	(a, b, c)	**15.**	(a, b, c)	**16.**	(a, b, c)	**17.**	(a, b, c, d)	**18.**	(a, c)
19.	(a, b, c)	**20.**	(a, d)	**21.**	(b, c)	**22.**	(a, b, c, d)	**23.**	(b, c)	**24.**	(a, b)
25.	(a, c)	**26.**	(a, b, c, d)	**27.**	(c, d)	**28.**	(a, b, c, d)	**29.**	(a, b, d)	**30.**	(a, c, d)
31.	(b, c)	**32.**	(a, b, d)	**33.**	(b, d)	**34.**	(a)	**35.**	(a, d)	**36.**	(a, c, d)

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $f(x) = \lim_{n\to\infty} n^2 \tan\left(\ln\left(\sec\frac{x}{n}\right)\right)$ and $g(x) = \min(f(x), \{x\})$

(where $\{\cdot\}$ denotes fractional part function)

1. Left hand derivative of $\phi(x) = e^{\sqrt{2f(x)}}$ at $x = 0$ is :

(a) 0 (b) 1 (c) -1 (d) Does not exist

2. Number of points in $x \in [-1, 2]$ at which $g(x)$ is discontinuous :

(a) 2 (b) 1 (c) 0 (d) 3

Paragraph for Question Nos. 3 to 4

Let $f(x)$ and $g(x)$ be two differentiable functions, defined as :

$f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$.

3. The value of $f(1) + g(-1)$ is :

(a) 0 (b) 1 (c) 2 (d) 3

4. The number of integers in the domain of the function $F(x) = \sqrt{-\frac{f(x)}{g(x)}} + \sqrt{3-x}$ is :

(a) 0 (b) 1 (c) 2 (d) Infinite

Paragraph for Question Nos. 5 to 6

Define : $f(x) = |x^2 - 4x + 3|\ln x + 2(x-2)^{1/3}, x > 0$

$$h(x) = \begin{cases} x-1 & , x \in Q \\ x^2 - x - 2 & , x \notin Q \end{cases}$$

5. $f(x)$ is non-differentiable at .... points and the sum of corresponding x value(s) is ....

(a) 3, 6 (b) 2, 3 (c) 2, 4 (d) 2, 5

6. $h(x)$ is discontinuous at $x =$......

(a) $1+\sqrt{2}$ (b) $\tan\frac{3\pi}{8}$ (c) $\tan\frac{7\pi}{8}$ (d) $\sqrt{2}-1$

Paragraph for Question Nos. 7 to 8

Consider a function defined in $[-2, 2]$

$$f(x) = \begin{cases} \{x\} & -2 \le x < -1 \\ |\operatorname{sgn} x| & -1 \le x \le 1 \\ \{-x\} & 1 < x \le 2 \end{cases}$$

where $\{\cdot\}$ denotes the fractional part function.

7. The total number of points of discontinuity of $f(x)$ for $x \in [-2, 2]$ is :

(a) 0 (b) 1 (c) 2 (d) 4

8. The number of points for $x \in [-2, 2]$ where $f(x)$ is non-differentiable is :

(a) 0 (b) 1 (c) 2 (d) 3

Paragraph for Question Nos. 9 to 10

Consider a function $f(x)$ in $[0, 2\pi]$ defined as :

$$f(x) = \begin{cases} [\sin x] + [\cos x] & ; \quad 0 \le x \le \pi \\ [\sin x] - [\cos x] & ; \quad \pi < x \le 2\pi \end{cases}$$

where $[\cdot]$ denotes greatest integer function then

9. Number of points where $f(x)$ is non-derivable :

(a) 2 (b) 3 (c) 4 (d) 5

10. $\lim\limits_{x \to \left(\frac{3\pi}{2}\right)^+} f(x)$ equals

(a) 0 (b) 1 (c) -1 (d) 2

Paragraph for Question Nos. 11 to 13

Let $f(x) = \begin{cases} x[x] & 0 \le x < 2 \\ (x-1)[x] & 2 \le x \le 3 \end{cases}$ where $[x]$ = greatest integer less than or equal to x, then :

11. The number of values of x for $x \in [0, 3]$ where $f(x)$ is discontinuous is :

(a) 0 (b) 1 (c) 2 (d) 3

12. The number of values of x for $x \in [0, 3]$ where $f(x)$ is non-differentiable is :

(a) 0 (b) 1 (c) 2 (d) 3

13. The number of integers in the range of $y = f(x)$ is :

(a) 3 (b) 4 (c) 5 (d) 6

Paragraph for Question Nos. 14 to 16

Let $f : R \to R$ be a continuous and differentiable function such that $f(x + y) = f(x) \cdot f(y)$ $\forall\, x, y, f(x) \ne 0$ and $f(0) = 1$ and $f'(0) = 2$.

Let $g(xy) = g(x) \cdot g(y)\ \forall\, x, y$ and $g'(1) = 2; g(1) \ne 0$

14. Identify the correct option :

(a) $f(2) = e^4$ (b) $f(2) = 2e^2$ (c) $f(1) < 4$ (d) $f(3) > 729$

15. Identify the correct option :

(a) $g(2) = 2$ (b) $g(3) = 3$ (c) $g(3) = 9$ (d) $g(3) = 6$

16. The number of values of x, where $f(x) = g(x)$:

(a) 0 (b) 1 (c) 2 (d) 3

Paragraph for Question Nos. 17 to 18

Let $f(x) = \dfrac{\cos^2 x}{1 + \cos x + \cos^2 x}$ and $g(x) = \lambda \tan x + (1 - \lambda)\sin x - x$, where $\lambda \in R$ and $x \in [0, \pi/2)$.

17. $g'(x)$ equals

(a) $\dfrac{(1 - \cos x)(f(x) - \lambda)}{\cos x}$ (b) $\dfrac{(1 - \cos x)(\lambda - f(x))}{\cos x}$

(c) $\dfrac{(1 - \cos x)(\lambda - f(x))}{f(x)}$ (d) $\dfrac{(1 - \cos x)(\lambda - f(x))}{(f(x))^2}$

18. The exhaustive set of values of 'λ' such that $g'(x) \geq 0$ for any $x \in [0, \pi/2)$:

(a) $[1, \infty)$ (b) $[0, \infty)$ (c) $\left[\dfrac{1}{2}, \infty\right)$ (d) $\left[\dfrac{1}{3}, \infty\right)$

Paragraph for Question Nos. 19 to 21

Let $f(x) = \lim\limits_{n \to \infty} \dfrac{x^2 + 2(x+1)^{2n}}{(x+1)^{2n+1} + x^2 + 1}, n \in N$ and

$g(x) = \tan\left(\dfrac{1}{2}\sin^{-1}\left(\dfrac{2f(x)}{1 + f^2(x)}\right)\right)$, then

19. The number of points where $g(x)$ is non-differentiable $\forall\, x \in R$ is :

(a) 1 (b) 2 (c) 3 (d) 4

20. $\lim\limits_{x \to -3} \dfrac{(x^2 + 4x + 3)}{\sin(x+3)\, g(x)}$ is equal to :

(a) 1 (b) 2 (c) 4 (d) Non-existent

21. $\lim\limits_{x \to 0^-} \left\{\dfrac{f(x)}{\tan^2 x}\right\} + \left|\lim\limits_{x \to -2^-} f(x)\right| + \lim\limits_{x \to -2^+} (5f(x))$ is equal to

(where $\{\cdot\}$ denotes fraction part function)

(a) 7 (b) 8 (c) 12 (d) Non-existent

Paragraph for Question Nos. 22 to 24

Let f and g be two differentiable functions such that :

$$f(x) = g'(1)\sin x + (g''(2) - 1)x$$

$$g(x) = x^2 - f'\left(\frac{\pi}{2}\right)x + f''\left(-\frac{\pi}{2}\right)$$

22. The number of solution(s) of the equation $f(x) = g(x)$ is/are :

(a) 1 (b) 2 (c) 3 (d) infinite

23. If $\int \frac{g(\cos x)}{f(x) - x}dx = \cos x + ln\ (h(x)) + C$ where C is constant and $h\left(\frac{\pi}{2}\right) = 1$ then $\left|h\left(\frac{2\pi}{3}\right)\right|$ is :

(a) $3\sqrt{2}$ (b) $2\sqrt{3}$ (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

24. If $\phi(x) = f^{-1}(x)$ then $\phi'\left(\frac{\pi}{2} + 1\right)$ equals to :

(a) $\frac{\pi}{2} + 1$ (b) $\frac{\pi}{2}$ (c) 1 (d) 0

Paragraph for Question Nos. 25 to 26

Suppose a function $f(x)$ satisfies the following conditions

$$f(x + y) = \frac{f(x) + f(y)}{1 + f(x)\,f(y)},\ \forall\ x, y \in R \text{ and } f'(0) = 1$$

Also $-1 < f(x) < 1,\ \forall\ x \in R$

25. $f(x)$ increases in the complete interval :

(a) $(-\infty, -1) \cup (-1, 0) \cup (0, 1) \cup (1, \infty)$ (b) $(-\infty, \infty)$

(c) $(-\infty, -1) \cup (-1, 0)$ (d) $(0, 1) \cup (1, \infty)$

26. The value of the limit $\underset{x \to \infty}{lt}\ (f(x))^x$ is :

(a) 0 (b) 1 (c) e (d) e^2

Paragraph for Question Nos. 27 to 28

Let $f(x)$ be a polynomial satisfying $\lim_{x \to \infty} \frac{x^4 f(x)}{x^8 + 1} = 3$

$f(2) = 5, f(3) = 10, f(-1) = 2, f(-6) = 37$

27. The value of $\lim\limits_{x\to -6} \dfrac{f(x)-x^2-1}{3(x+6)}$ equals to :

(a) $-\lfloor 6$ (b) $\lfloor 6$ (c) $\dfrac{\lfloor 6}{2}$ (d) $\dfrac{-\lfloor 6}{2}$

28. The number of points of discontinuity of $g(x) = \dfrac{1}{x^2+1-f(x)}$ in $\left[\dfrac{-15}{2}, \dfrac{5}{2}\right]$ equals :

(a) 4 (b) 3 (c) 1 (d) 0

Paragraph for Question Nos. 29 to 30

Consider $f(x) = x^{\ln x}$ and $g(x) = e^2 x$. Let α and β be two values of x satisfying $f(x) = g(x)$ $(\alpha < \beta)$

29. If $\lim\limits_{x\to\beta} \dfrac{f(x)-c\beta}{g(x)-\beta^2} = l$ then the value of $c - l$ equals to :

(a) $4-e^2$ (b) e^2-4 (c) $4-e$ (d) $e-4$

30. If $h(x) = \dfrac{f(x)}{g(x)}$ then $h'(\alpha)$ equals to :

(a) e (b) $-e$ (c) $3e$ (d) $-3e$

Paragraph for Question Nos. 31 to 32

Let $f_n(x) + f_n(y) = \dfrac{x^n + y^n}{x^n y^n}$ $\forall\ x, y \in R - \{0\}$ where $n \in N$ and

$g(x) = \max\limits_{x} \left\{ f_2(x), f_3(x), \dfrac{1}{2} \right\}$ $\forall\ x \in R - \{0\}$

31. The minimum value of $\sum\limits_{k=1}^{\infty} f_{2k}(\operatorname{cosec}\theta) + \sum\limits_{k=1}^{\infty} f_{2k}(\sec\theta)$, where $\theta \neq \dfrac{k\pi}{2}$; $k \in I$ is :

(a) 1 (b) 2 (c) $\sqrt{2}$ (d) 4

32. The number of values of x for which $g(x)$ is non-differentiable $(x \in R - \{0\})$:

(a) 3 (b) 4 (c) 5 (d) 1

Answers

1.	(c)	**2.**	(a)	**3.**	(d)	**4.**	(c)	**5.**	(d)	**6.**	(d)	**7.**	(b)	**8.**	(d)	**9.**	(c)	**10.**	(c)
11.	(c)	**12.**	(d)	**13.**	(c)	**14.**	(a)	**15.**	(c)	**16.**	(b)	**17.**	(c)	**18.**	(d)	**19.**	(d)	**20.**	(b)
21.	(a)	**22.**	(b)	**23.**	(b)	**24.**	(c)	**25.**	(b)	**26.**	(b)	**27.**	(d)	**28.**	(b)	**29.**	(b)	**30.**	(d)
31.	(b)	**32.**	(a)																

Exercise-4 : Matching Type Problems

1.

	Column-I		Column-II
(A)	If $\int_0^{\pi} \frac{\log \sin x}{\cos^2 x} dx = -K$ then the value of $\frac{3k}{\pi}$ is greater than	**(P)**	0
(B)	If $e^{x+y} + e^{y-x} = 1$ and $y'' - (y')^2 + K = 0$, then K is equal to	**(Q)**	1
(C)	If $f(x) = x \ln x$ then $2(f^{-1})'(\ln 4)$ is more than	**(R)**	2
(D)	$\lim_{x \to \infty} (x \ln x)^{\frac{1}{x^2+1}}$ is less than	**(S)**	4
		(T)	5

2. Let $f(x) = \begin{cases} [x] & , & -2 \le x < 0 \\ |x| & , & 0 \le x \le 2 \end{cases}$

(where [·] denotes the greatest integer function) $g(x) = \sec x$, $x \in R - (2n+1)\frac{\pi}{2}, n \in I$

Match the following statements in column I with their values in column II in the interval $\left(-\frac{3\pi}{2}, \frac{3\pi}{2}\right)$.

	Column-I		Column-II
(A)	Abscissa of points where limit of $fog(x)$ exist is/are	**(P)**	-1
(B)	Abscissa of points in domain of $gof(x)$, where limit of $gof(x)$ does not exist is/are	**(Q)**	π
(C)	Abscissa of points of discontinuity of $fog(x)$ is/are	**(R)**	$\frac{5\pi}{6}$
(D)	Abscissa of points of differentiability of $fog(x)$ is/are	**(S)**	$-\pi$
		(T)	0

3. Let a function $f(x) = [x]\{x\} - |x|$ where [·], {·} are greatest integer and fractional part respectively then match the following List-I with List-II.

	Column-I		Column-II
(A)	$f(x)$ is continuous at x equal to	**(P)**	3
(B)	$\frac{4}{3}\left\vert\int_2^3 f(x)\,dx\right\vert$ is equal to	**(Q)**	1

(C)	If $g(x) = x - 1$ and if $f(x) = g(x)$ where $x \in (-3, \infty)$, then number of solutions	**(R)**	4
(D)	If $l = \lim_{x \to 4^+} f(x)$, then $-l$ is equal to	**(S)**	2

4.

	Column-I		Column-II
(A)	$\lim_{x\to\infty}\left(\dfrac{x^2+2x-1}{2x^2-3x-2}\right)^{\frac{2x+1}{2x-1}} =$	**(P)**	$\dfrac{1}{2}$
(B)	$\lim_{x\to 0}\dfrac{\log_{\sec x/2}\cos x}{\log_{\sec x}\cos\dfrac{x}{2}} =$	**(Q)**	2
(C)	Let $f(x) = \max.(\cos x, x, 2x-1)$ where $x \geq 0$ then number of points of non-differentiability of $f(x)$ is	**(R)**	5
(D)	If $f(x) = [2 + 3\sin x]$, $0 < x < \pi$ then number of points at which the function is discontinuous, is	**(S)**	16

5. The function
$$f(x) = \begin{cases} ax(x-1)+b & x < 1 \\ = x-1 & 1 \leq x \leq 3 \\ = px^2+qx+2 & x > 3 \end{cases}$$

if (i) $f(x)$ is continuous for all x

(ii) $f'(1)$ does not exist

(iii) $f'(x)$ is continuous at $x = 3$, then

	Column-I		Column-II
(A)	a cannot has value	**(P)**	1/3
(B)	b has value	**(Q)**	0
(C)	p has value	**(R)**	–1
(D)	q has value	**(S)**	1

Answers

1.	$A \to P, Q, R$; $B \to Q$; $C \to P, Q$; $D \to R, S, T$
2.	$A \to P, Q, R, S, T$; $B \to P, T$; $C \to Q, S$; $D \to P, R, T$
3.	$A \to Q$; $B \to S$; $C \to P$; $D \to R$
4.	$A \to P$; $B \to S$; $C \to Q$; $D \to R$
5.	$A \to S$; $B \to Q$; $C \to P$; $D \to R$

Exercise-5 : Subjective Type Problems

1. Let $f(x)=\begin{cases} ax(x-1)+b & ; \quad x<1 \\ x+2 & ; \quad 1\le x\le 3 \\ px^2+qx+2 & ; \quad x>3 \end{cases}$ is continuous $\forall x \in R$ except $x=1$ but $|f(x)|$ is differentiable everywhere and $f'(x)$ is continuous at $x=3$ and $|a+p+b+q|=k$, then $k=$

2. If $y=\sin(8\sin^{-1}x)$ then $(1-x^2)\dfrac{d^2y}{dx^2}-x\dfrac{dy}{dx}=-ky$, where $k=$

3. If $y^2=4ax$, then $\dfrac{d^2y}{dx^2}=\dfrac{ka^2}{y^3}$, where $k^2=$

4. The number of values of x, $x\in[-2,3]$ where $f(x)=[x^2]\sin(\pi x)$ is discontinuous is (where $[\cdot]$ denotes greatest integer function)

5. If $f(x)$ is continuous and differentiable in $[-3,9]$ and $f'(x)\in[-2,8]\ \forall x\in(-3,9)$. Let N be the number of divisors of the greatest possible value of $f(9)-f(-3)$, then find the sum of digits of N.

6. If $f(x)=\begin{cases} \cos x^3 & ; \quad x<0 \\ \sin x^3-|x^3-1| & ; \quad x\ge 0 \end{cases}$

then find the number of points where $g(x)=f(|x|)$ is non-differentiable.

7. Let $f(x)=x^2+ax+3$ and $g(x)=x+b$, where $F(x)=\lim\limits_{n\to\infty}\dfrac{f(x)+(x^2)^n g(x)}{1+(x^2)^n}$. If $F(x)$ is continuous at $x=1$ and $x=-1$ then find the value of (a^2+b^2).

8. Let $f(x)=\begin{cases} 2-x & , \quad -3\le x\le 0 \\ x-2 & , \quad 0<x<4 \end{cases}$

Then $f^{-1}(x)$ is discontinuous at $x=$

9. If $f(x)+2f(1-x)=x^2+2\ \forall x\in R$ and $f(x)$ is a differentiable function, then the value of $f'(8)$ is

10. Let $f(x)=signum(x)$ and $g(x)=x(x^2-10x+21)$, then the number of points of discontinuity of $f[g(x)]$ is

11. If $\dfrac{d^2}{dx^2}\left(\dfrac{\sin^4x+\sin^2x+1}{\sin^2x+\sin x+1}\right)=a\sin^2x+b\sin x+c$ then the value of $b+c-a$ is

12. If $f(x)=a\cos(\pi x)+b$, $f'\left(\dfrac{1}{2}\right)=\pi$ and $\displaystyle\int_{1/2}^{3/2} f(x)\,dx=\frac{2}{\pi}+1$, then find the value of $-\dfrac{12}{\pi}\left(\dfrac{\sin^{-1}a}{3}+\cos^{-1}b\right)$.

13. Let $\alpha(x) = f(x) - f(2x)$ and $\beta(x) = f(x) - f(4x)$

and $\alpha'(1) = 5\ \alpha'(2) = 7$

then find the value of $\beta'(1) - 10$

14. Let $f(x) = -4 \cdot e^{\frac{1-x}{2}} + \frac{x^3}{3} + \frac{x^2}{2} + x + 1$ and g be inverse function of f and $h(x) = \frac{a + bx^{3/2}}{x^{5/4}}$,

$h'(5) = 0$, then $\dfrac{a^2}{5b^2 g'\left(\frac{-7}{6}\right)} =$

15. If $y = e^{2\sin^{-1} x}$ then $\left|\dfrac{(x^2 - 1)y'' + xy'}{y}\right|$ is equal to

16. Let f be a continuous function on $[0, \infty)$ such that $\lim\limits_{x \to \infty}\left(f(x) + \int_0^x f(t)\,dt\right)$ exists. Find $\lim\limits_{x \to \infty} f(x)$.

17. Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and let $g(x) = f^{-1}(x)$. Find $g'''(0)$.

18. If $f(x) = \begin{cases} \cos x^3 & ; \ x < 0 \\ \sin x^3 - |x^3 - 1| & ; \ x \geq 0 \end{cases}$

then find the number of points where $g(x) = f(|x|)$ is non-differentiable.

19. Let $f: R^+ \longrightarrow R$ be a differentiable function satisfying :

$f(xy) = \dfrac{f(x)}{y} + \dfrac{f(y)}{x}\ \forall x, y \in R^+$ also $f(1) = 0; f'(1) = 1$

find $\lim\limits_{x \to e}\left[\dfrac{1}{f(x)}\right]$ (where $[\cdot]$ denotes greatest integer function).

20. For the curve $\sin x + \sin y = 1$ lying in the first quadrant there exists a constant α for which $\lim\limits_{x \to 0} x^{\alpha}\dfrac{d^2y}{dx^2} = L$ (not zero), then $2\alpha =$

21. Let $f(x) = x\tan^{-1}(x^2) + x^4$. Let $f^k(x)$ denotes k^{th} derivative of $f(x)$ w.r.t. x, $k \in N$. If $f^{2m}(0) \neq 0, m \in N$, then $m =$

22. If $x = \cos\theta$ and $y = \sin^3\theta$, then $\left|\dfrac{yd^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2\right|$ at $\theta = \dfrac{\pi}{2}$ is :

23. The value of x, $x \in (2, \infty)$ where $f(x) = \sqrt{x + \sqrt{8x - 16}} + \sqrt{x - \sqrt{8x - 16}}$ is not differentiable is :

24. The number of non differentiability points of function $f(x) = \min\left([x], \{x\}, \left|x - \frac{3}{2}\right|\right)$ for $x \in (0, 2)$, where $[\cdot]$ and $\{\cdot\}$ denote greatest integer function and fractional part function respectively.

Answers

1.	3	**2.**	64	**3.**	16	**4.**	8	**5.**	3	**6.**	2	**7.**	17
8.	2	**9.**	4	**10.**	3	**11.**	7	**12.**	2	**13.**	9	**14.**	5
15.	4	**16.**	0	**17.**	1	**18.**	2	**19.**	2	**20.**	3	**21.**	2
22.	3	**23.**	4	**24.**	3								

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4 APPLICATION OF DERIVATIVES

Exercise-1 : Single Choice Problems

1. The difference between the maximum and minimum value of the function $f(x) = 3\sin^4 x - \cos^6 x$ is :

(a) $\frac{3}{2}$ (b) $\frac{5}{2}$ (c) 3 (d) 4

2. A function $y = f(x)$ has a second order derivative $f''(x) = 6(x-1)$. If its graph passes through the point (2, 1) and at that point the tangent to the graph is $y = 3x - 5$, then the function is :

(a) $(x-1)^2$ (b) $(x-1)^3$ (c) $(x+1)^3$ (d) $(x+1)^2$

3. If the subnormal at any point on the curve $y = 3^{1-k} \cdot x^k$ is of constant length then k equals to :

(a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 0

4. If $x^5 - 5qx + 4r$ is divisible by $(x-c)^2$ then which of the following must hold true $\forall\, q, r, c \in R$?

(a) $q = r$ (b) $q + r = 0$ (c) $q^5 = r^4$ (d) $q^4 = r^5$

5. A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50\,\text{cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate at which the thickness of ice decreases, is :

(a) $\frac{1}{36\pi}$ cm/min (b) $\frac{1}{18\pi}$ cm/min (c) $\frac{1}{54\pi}$ cm/min (d) $\frac{5}{6\pi}$ cm/min

6. If $f(x) = \frac{(x-1)(x-2)}{(x-3)(x-4)}$, then number of local extremas for $g(x)$, where $g(x) = f(|x|)$:

(a) 3 (b) 4 (c) 5 (d) None of these

7. Two straight roads OA and OB intersect at an angle 60°. A car approaches O from A, where $OA = 700\,\text{m}$ at a uniform speed of 20 m/s, Simultaneously, a runner starts running from O towards B at a uniform speed of 5 m/s. The time after start when the car and the runner are closest is :

(a) 10 sec (b) 15 sec

(c) 20 sec (d) 30 sec

8. Let $f(x)=\begin{cases} a-3x & ; & -2\le x<0 \\ 4x+3 & ; & 0\le x<1 \end{cases}$; if $f(x)$ has smallest value at $x=0$, then range of a, is :

(a) $(-\infty,3)$ (b) $(-\infty,3]$ (c) $(3,\infty)$ (d) $[3,\infty)$

9. $f(x)=\begin{cases} 3+|x-k| & , & x\le k \\ a^2-2+\dfrac{\sin(x-k)}{(x-k)} & , & x>k \end{cases}$ has minimum at $x=k$, then :

(a) $a\in R$ (b) $|a|<2$ (c) $|a|>2$ (d) $1<|a|<2$

10. For a certain curve $\dfrac{d^2y}{dx^2}=6x-4$ and curve has local minimum value 5 at $x=1$. Let the global maximum and global minimum values, where $0\le x\le 2$; are M and m. Then the value of $(M-m)$ equals to :

(a) -2 (b) 2 (c) 12 (d) -12

11. The tangent to $y=ax^2+bx+\dfrac{7}{2}$ at (1, 2) is parallel to the normal at the point (–2, 2) on the curve $y=x^2+6x+10$. Then the value of $\dfrac{a}{2}-b$ is :

(a) 2 (b) 0 (c) 3 (d) 1

12. If (a,b) be the point on the curve $9y^2=x^3$ where normal to the curve make equal intercepts with the axis, then the value of $(a+b)$ is :

(a) 0 (b) $\dfrac{10}{3}$ (c) $\dfrac{20}{3}$ (d) None of these

13. The curve $y=f(x)$ satisfies $\dfrac{d^2y}{dx^2}=6x-4$ and $f(x)$ has a local minimum value 5 when $x=1$. Then $f(0)$ is equal to :

(a) 1 (b) 0 (c) 5 (d) None of these

14. Let A be the point where the curve $5\alpha^2x^3+10\alpha x^2+x+2y-4=0$ $(\alpha\in R,\alpha\ne 0)$ meets the y-axis, then the equation of tangent to the curve at the point where normal at A meets the curve again, is :

(a) $x-\alpha y+2\alpha=0$ (b) $\alpha x+y-2=0$ (c) $2x-y+2=0$ (d) $x+2y-4=0$

15. The difference between the greatest and the least value of the function $f(x)=\cos x+\dfrac{1}{2}\cos 2x-\dfrac{1}{3}\cos 3x$

(a) $\dfrac{11}{5}$ (b) $\dfrac{13}{6}$ (c) $\dfrac{9}{4}$ (d) $\dfrac{7}{3}$

16. The x co-ordinate of the point on the curve $y=\sqrt{x}$ which is closest to the point (2, 1) is :

(a) $\dfrac{2+\sqrt{3}}{2}$ (b) $\dfrac{1+\sqrt{3}}{2}$ (c) $\dfrac{-1+\sqrt{3}}{2}$ (d) 1

17. The tangent at a point P on the curve $y = \ln\left(\frac{2+\sqrt{4-x^2}}{2-\sqrt{4-x^2}}\right) - \sqrt{4-x^2}$ meets the y-axis at T; then PT^2 equals to :

(a) 2 (b) 4 (c) 8 (d) 16

18. Let $f(x) = \int_{x^2}^{x^3} \frac{dt}{\ln t}$ for $x > 1$

and $g(x) = \int_1^x (2t^2 - \ln t) f(t)\, dt$ $(x > 1)$, then :

(a) g is increasing on $(1, \infty)$
(b) g is decreasing on $(1, \infty)$
(c) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$
(d) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

19. Let $f(x) = x^3 + 6x^2 + ax + 2$, if $(-3, -1)$ is the largest possible interval for which $f(x)$ is decreasing function, then $a =$

(a) 3 (b) 9 (c) -2 (d) 1

20. Let $f(x) = \tan^{-1}\left(\frac{1-x}{1+x}\right)$. Then difference of the greatest and least value of $f(x)$ on $[0, 1]$ is :

(a) $\pi/2$ (b) $\pi/4$ (c) π (d) $\pi/3$

21. The number of integral values of a for which $f(x) = x^3 + (a+2)x^2 + 3ax + 5$ is monotonic in $\forall x \in R$.

(a) 2 (b) 4 (c) 6 (d) 7

22. The number of critical points of $f(x) = \left(\int_0^x (\cos^2 t - \sqrt[3]{t})\, dt\right) + \frac{3}{4}x^{4/3} - \frac{x+1}{2}$ in $[0, 6\pi]$ is :

(a) 10 (b) 8 (c) 6 (d) 12

23. Let $f(x) = \min\left(\frac{1}{2} - \frac{3x^2}{4}, \frac{5x^2}{4}\right)$ for $0 \le x \le 1$, then maximum value of $f(x)$ is :

(a) 0 (b) $\frac{5}{64}$

(c) $\frac{5}{4}$ (d) $\frac{5}{16}$

24. Let $f(x) = \begin{cases} 2 - |x^2 + 5x + 6| & x \ne -2 \\ b^2 + 1 & x = -2 \end{cases}$

Has relative maximum at $x = -2$, then complete set of values b can take is :

(a) $|b| \ge 1$ (b) $|b| < 1$ (c) $b > 1$ (d) $b < 1$

25. Let for the function $f(x) = \begin{cases} \cos^{-1} x & ; & -1 \le x \le 0 \\ mx + c & ; & 0 < x \le 1 \end{cases}$;

Lagrange's mean value theorem is applicable in $[-1, 1]$ then ordered pair (m, c) is :

(a) $\left(1, -\frac{\pi}{2}\right)$ (b) $\left(1, \frac{\pi}{2}\right)$ (c) $\left(-1, -\frac{\pi}{2}\right)$ (d) $\left(-1, \frac{\pi}{2}\right)$

26. Tangents are drawn to $y = \cos x$ from origin then points of contact of these tangents will always lie on :

(a) $\frac{1}{x^2} = \frac{1}{y^2} + 1$ (b) $\frac{1}{x^2} = \frac{1}{y^2} - 2$ (c) $\frac{1}{y^2} = \frac{1}{x^2} + 1$ (d) $\frac{1}{y^2} = \frac{1}{x^2} - 2$

27. Least natural number a for which $x + ax^{-2} > 2 \; \forall \; x \in (0, \infty)$ is :

(a) 1 (b) 2 (c) 5 (d) None of these

28. Angle between the tangents to the curve $y = x^2 - 5x + 6$ at points (2, 0) and (3, 0) is :

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

29. Difference between the greatest and least values of the function $f(x) = \int_0^x (\cos^2 t + \cos t + 2)\, dt$ in the interval $[0, 2\pi]$ is $K\pi$, then K is equal to :

(a) 1 (b) 3 (c) 5 (d) None of these

30. The range of the function $f(\theta) = \frac{\sin\theta}{\theta} + \frac{\theta}{\tan\theta}, \theta \in \left(0, \frac{\pi}{2}\right)$ is equal to :

(a) $(0, \infty)$ (b) $\left(\frac{1}{\pi}, 2\right)$ (c) $(2, \infty)$ (d) $\left(\frac{2}{\pi}, 2\right)$

31. Number of integers in the range of c so that the equation $x^3 - 3x + c = 0$ has all its roots real and distinct is :

(a) 2 (b) 3 (c) 4 (d) 5

32. Let $f(x) = \int e^x (x-1)(x-2)\, dx$. Then $f(x)$ decreases in the interval :

(a) $(2, \infty)$ (b) $(-2, -1)$
(c) $(1, 2)$ (d) $(-\infty, 1) \cup (2, \infty)$

33. If the cubic polynomial $y = ax^3 + bx^2 + cx + d \; (a, b, c, d \in R)$ has only one critical point in its entire domain and $ac = 2$, then the value of $|b|$ is :

(a) $\sqrt{2}$ (b) $\sqrt{3}$ (c) $\sqrt{5}$ (d) $\sqrt{6}$

34. On the curve $y = \frac{1}{1+x^2}$, the point at which $\left|\frac{dy}{dx}\right|$ is greatest in the first quadrant is :

(a) $\left(\frac{1}{2}, \frac{4}{5}\right)$ (b) $\left(1, \frac{1}{2}\right)$ (c) $\left(\frac{1}{\sqrt{2}}, \frac{2}{3}\right)$ (d) $\left(\frac{1}{\sqrt{3}}, \frac{3}{4}\right)$